Hyperclosure of the Odd Six-Dimensional Box Integrals

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Abstract

For $n \geq 1$ write $B_n(s) = \int_{[0,1]^n} (x_1^2 + \cdots + x_n^2)^{s/2} \, \mathrm{d}^n x$. The outstanding six-dimensional question highlighted by Borwein–Chan–Crandall (2010) asks for hyperclosed expressions for odd-integer moments, notably $B_6(1)$, the mean distance from a uniform point of the six-cube to a corner. Their Theorem 7.2 reduces the problem to $B_6(-7)$, $B_6(-5)$, and $B_6(-3)$. We provide a finite classical-polylogarithmic formula for the central seed $B_6(-5)$ through 1,677 trilogarithm terms, 24 logarithm–dilogarithm terms, and 1,414 cubic logarithm monomials, all specified by literal tables. The algebraic certificate comprises exact characteristic-zero checks of ramified rational-map fibers and all $24^3 = 13,824$ word-tensor coefficients. We then prove a dilogarithmic formula for $B_6(-7)$, eliminate $B_6(-3)$ by the original box recurrences, and give an explicit formula for $B_6(1)$. In particular, the combined construction establishes hyperclosure of all integer six-dimensional box integrals except $s = -6$, where the meromorphic continuation has a simple pole. Independent numerical tests include a second algebraic endpoint and a direct Laplace evaluation of $B_6(1)$.

1 Introduction: the six-dimensional box-integral problem

For a complex parameter s with $\Re s > -n$, define the *box integral*

$$B_n(s) = \int_{[0,1]^n} \left(x_1^2 + \cdots + x_n^2\right)^{s/2} \mathrm{d}x_1 \cdots \mathrm{d}x_n. \tag{1}$$

The restriction ensures absolute convergence at the origin. These integrals are moments of the distance of a uniformly sampled point of the unit cube from one of its corners. In particular, the original geometric quantity of interest is

$$\boxed{B_6(1) = \int_{[0,1]^6} \sqrt{x_1^2 + \cdots + x_6^2} \, \mathrm{d}^6 x.} \tag{2}$$

It is important not to confuse this single-point distance with the expected distance between *two* independent points of the cube, which is a different box-integral family.

The defining integral is finite only for $\Re s > -6$. As shown by Borwein–Chan–Crandall [2, §3], $B_6(s)$ extends meromorphically to the complex plane with a unique simple pole at $s = -6$, whose residue is $\pi^3/64$.

An evaluation is *hyperclosed* in the sense of Bailey–Borwein–Crandall and Borwein–Chan–Crandall [1, 2] if it lies in the ring generated, under finite sums and products, by generalized-hypergeometric evaluations with algebraic arguments and rational hypergeometric coefficient recurrences. Write $\mathcal{H}$ for this ring. This makes "closed form" a precise algebraic criterion; arbitrary unevaluated integrals or infinite collections of special values do not qualify. Classical logarithms, dilogarithms and trilogarithms at algebraic arguments belong to this ring.

In dimension six, Borwein–Chan–Crandall [2, §7, Theorems 7.1–7.2] proved that all even-index $B_6(2k)$ are hyperclosed, and that the three exceptional values

$$\boxed{B_6(-7), \qquad B_6(-5), \qquad B_6(-3)} \tag{3}$$

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suffice for all odd-index values, including the original $B_6(1)$. These negative exponents arise naturally from the recurrences; $B_6(-7)$ means the meromorphic continuation, since its naive six-dimensional defining integral diverges.

The decisive new piece is a finite evaluation of $B_6(-5)$. The remaining two seeds are dispatched in Section 6: $B_6(-7)$ has a short explicit dilogarithmic expression, while $B_6(-3)$ is reduced to $B_6(-5)$, $B_6(-7)$ and one explicitly dilogarithmic shifted three-dimensional integral. Thus the cumulative conclusion is *hyperclosure at every integer parameter other than the pole* $s = -6$, not merely of a single seed. The full exact formula is lengthy, so a short explicit identity for $B_6(1)$ is also displayed below, referring only to four finite, fully specified special-value expressions.

Main contribution. Theorem 1 proves

$$\boxed{B_6(s) \in \mathcal{H} \quad (s \in \mathbb{Z} \setminus \{-6\}).} \tag{4}$$

The decisive formula, equation (18), gives a fully indexed finite combination of classical Li_2 and Li_3 values and logarithms at algebraic arguments, together with finite products of these values. The value is

$$B_6(-5) = 0.70227312325610135940870498837302070 1\ldots.$$

The unusual length of the formula is a feature of the construction, not a claim of intrinsic complexity. The proof is exact: the central certificate is an equality of *full* length-three tensors rather than an identity only modulo products or a numerical relation.

Statement for the motivating integral. Put

$$L = \log \frac{1+\sqrt{6}}{\sqrt{5}}, \qquad \theta = \arctan \sqrt{\frac{2}{3}}.$$

Writing $D = C_{3,0}(-3,3)$ for the explicitly evaluated dilogarithmic constant in Section 6, we prove

$$\boxed{B_6(1) = \frac{B_6(-5)}{112} - \frac{67B_6(-7)}{168} - \frac{133D}{42} + 5L + \frac{60}{7}\theta - \frac{15\pi}{7} + \frac{\sqrt{6}}{7}.} \tag{5}$$

Here $B_6(-5)$ is given by the complete 3,115-entry finite formula in Theorem 1 and its appendix; $B_6(-7)$ and D are specified by finite dilogarithm expressions in Section 6. Thus (5) contains no unevaluated integrals. An independent, cancellation-stabilized Laplace benchmark is

$$B_6(1) = 1.38857408445734784253025407303078881591\ldots.$$

Guide to the proof. Section 2 derives the reduction from the original six-dimensional integral and introduces three logarithmic one-forms on an elliptic curve. Section 3 constructs the finite algebraic alphabet and its degree-two and degree-three rational maps. Section 4 defines the normalized classical-polylogarithmic primitives and states the exact evaluation. Section 5 integrates the certified tensor identity using Chen's shuffle relations; Section 6 proves the remaining seeds and the explicit formula for $B_6(1)$; Section 7 records independent numerical comparisons. The appendix contains *all* 3,115 literal coefficient entries and all algebraic data needed to reconstruct the formula and check the finite tensor identity within the paper.

2 From the six-dimensional integral to elliptic logarithms

The key analytic reduction is the following exact ordered integral; its derivation from (1) is supplied immediately below:

$$B_6(-5) = 60 I_{abc}(1/3), \qquad a = \mathrm{d}\arcsin\frac{t}{1+2t}, \quad b = \mathrm{d}\arcsin t, \quad c = \mathrm{d}\,\mathrm{artanh}\sqrt{\frac{t}{1-t}}, \tag{6}$$

where $I_{abc}(T) = \int_{0<t_1<t_2<t_3<T} a(t_1)b(t_2)c(t_3)$; the earliest differential is the first letter. An elementary rationalizing substitution for the first two radicals is

$$N(u) = u^4 - 12u^2 + 4, \quad D(u) = u^4 + 20u^2 + 4, \quad t = \frac{N(u)}{D(u)}.$$

The physical arc γ on $\mathcal{E} : Y^2 = N(u)$ runs, with $Y \geq 0$, from

$$q_0 = (2+\sqrt{2}, 0) \quad \text{to} \quad q_1 = (2\sqrt{2}+\sqrt{6},\, 4(2\sqrt{2}+\sqrt{6})).$$

Set $\alpha = (2 + i\sqrt{2})/\sqrt{3}$ and introduce rational functions

$$f = \frac{2\sqrt{2}+i}{3}\frac{u^2-\alpha^2}{u^2-\bar{\alpha}^2}, \qquad g = i\frac{u^2-4iu+2}{u^2+4iu+2}, \qquad h = \frac{4\sqrt{2}u+Y}{4\sqrt{2}u-Y}. \tag{7}$$

Along γ the normalized logarithms satisfy $\log f = i\arcsin(t/(1+2t))$, $\log g = i\arcsin t$, $\log h = 2\,\mathrm{artanh}\sqrt{t/(1-t)}$, and $f(q_0) = g(q_0) = h(q_0) = 1$. Therefore

$$\boxed{B_6(-5) = -30 I_{fgh}(\gamma)}, \qquad I_{fgh} = \int_{0<s<t<v<1} \gamma^*\,\mathrm{d}\log f(s)\;\gamma^*\,\mathrm{d}\log g(t)\;\gamma^*\,\mathrm{d}\log h(v). \tag{8}$$

The quartic curve $\mathcal{E}$ is genus one and is birational to the elliptic curve $E : y^2 = x^3 - 4x$ introduced in the next section. The latter has $j(E) = 1728$, so it has complex multiplication over $\overline{\mathbb{Q}}$. This elliptic description is an intermediate organizing device: the final answer will involve only classical polylogarithms at algebraic arguments.

2.1 Derivation of the six-dimensional reduction

The factor in (6) can be derived independently, without assuming any of the algebraic-divisor certificates. By partitioning the unit cube according to its largest coordinate and using homogeneity,

$$B_6(-5) = 6\int_{[0,1]^5}(1+\sum_{i=1}^{5}x_i^2)^{-5/2}\,\mathrm{d}^5x = \frac{\pi^2}{4}\int_0^\infty e^{-a}\frac{\mathrm{erf}(\sqrt{a})^5}{a}\,\mathrm{d}a.$$

Define $H(s) = \int_0^\infty e^{-sa}\,\mathrm{erf}(\sqrt{a})^5\;\mathrm{d}a/a$. Integration by parts and the substitution $z = \sqrt{a}$ give

$$H'(s) = -\frac{5}{s\sqrt{s+1}}\,M_4((s+2)^{-1}),$$

where $M_4(\rho) = \mathbb{E}[\prod_{j=1}^4 \operatorname{sgn} X_j]$ for four standard jointly Gaussian variables with common off-diagonal correlation ρ. To see the normalization, condition on a common Gaussian factor; then $M_4((s+2)^{-1})$ equals the fourth moment of the corresponding error-function transform. Gaussian covariance differentiation, applied to the six coordinate pairs, and the conditional bivariate arcsine law yield

$$M_4(0) = 0, \qquad M_4'(\rho) = \frac{24}{\pi^2\sqrt{1-\rho^2}}\arcsin\!\left(\frac{\rho}{1+2\rho}\right).$$

Indeed the density of any selected pair at zero is $(2\pi\sqrt{1-\rho^2})^{-1}$, Price differentiation contributes a factor 4, and the remaining conditional sign-correlation is $(2/\pi)\arcsin(\rho/(1+2\rho))$. Writing $F(t) = \int_0^t \arcsin(u/(1+2u))\,\mathrm{d}u/\sqrt{1-u^2}$, integrating H' from $s=1$ to infinity, and setting $t=(s+2)^{-1}$, gives

$$B_6(-5) = 30\int_0^{1/3} \frac{F(t)}{(1-2t)\sqrt{t(1-t)}}\,\mathrm{d}t.$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{artanh}\sqrt{\frac{t}{1-t}} = \frac{1}{2(1-2t)\sqrt{t(1-t)}},$$

Fubini's theorem proves (6). (The original integrands are nonnegative, so all interchanges also follow from Tonelli's theorem.)

The logarithmic uniformization in (7) can be checked by finite polynomial algebra: for $t = N(u)/D(u)$ one has

$$\frac{f-f^{-1}}{2i} = \frac{t}{1+2t}, \qquad \frac{g-g^{-1}}{2i} = t, \qquad \frac{h-1}{h+1} = \frac{Y}{4\sqrt{2}\,u} = \sqrt{\frac{t}{1-t}}.$$

The initial values of f, g, h are all 1, and the arguments vary continuously on the real arc. These identities establish the logarithmic normalizations and show $a = -i\,\mathrm{d}\log f$, $b = -i\,\mathrm{d}\log g$, $c = \frac{1}{2}\,\mathrm{d}\log h$ as required.

3 The marked elliptic alphabet and its rational maps

The logarithmic differentials in the preceding section have zeros and poles at just 24 algebraic points. These points provide the finite *alphabet* for the tensor computation. To express a length-three word in classical polylogarithms, we construct rational maps of degrees two and three whose 0, 1, and ∞ fibers are supported on this alphabet. The following coordinates and normalizations make that construction explicit.

Write $E : y^2 = x^3 - 4x$. The birational map from the quartic $\mathcal{E}$ to the Weierstrass model E is

$$x = \frac{Y+2}{u^2} - 1, \qquad y = \frac{2(Y+2-3u^2)}{u^3}, \qquad u = \frac{2(x-2)}{y}. \tag{9}$$

Use the degree-16 number field $\mathsf{K} = \mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5})$. We label the eight zeros and poles of each of f, g, h by $\epsilon, \delta, \eta \in \{\pm 1\}$:

family	u	Y
f	$\delta\dfrac{2+\epsilon i\sqrt{2}}{\sqrt{3}}$	$\eta\dfrac{8}{3}(-\epsilon + i\sqrt{2})$
g	$\delta i(2+\epsilon\sqrt{6})$	$8\eta(\sqrt{3}+\epsilon\sqrt{2})$
h	$\delta(2\sqrt{3}+\epsilon\sqrt{10})$	$4\sqrt{2}\eta u$

(10)

The resulting 24 Weierstrass points $P_0, \ldots, P_{23}$ are ordered lexicographically by the tuple `(family,epsilon,delta,eta)`, with $f < g < h$ and $-1 < 1$; their precise labels are reproduced in Appendix A.1. They are distinct and avoid the 2-torsion $y = 0$.

We define an entirely algebraic rational function from every row j of the complete coefficient-and-fiber list in Appendix A.4. That list records three pairwise-disjoint effective divisors Z_j, K_j, O_j of degree 2 or 3 on the 24 points, intended as the fibers above $0, \infty, 1$.

Let $\mathcal{O}$ be the Weierstrass origin. For points $P, Q \in E(\mathsf{K})$ write

$$g_{P,Q} = \frac{\ell_{P,Q}}{v_{P+Q}}, \qquad v_T(X,Y) = X - x(T), \qquad g_{P,-P} = v_P, \tag{11}$$

where $\ell_{P,Q}$ is the secant or tangent, and the conventional factor involving $\mathcal{O}$ is 1. For an effective ordered divisor $D = P + Q$ or $P + Q + R$ set

$$M_{P+Q} = g_{P,Q}, \qquad M_{P+Q+R} = g_{P,Q} g_{P+Q,R}. \tag{12}$$

The second formula remains valid when $P + Q = \mathcal{O}$, by interpreting $g_{\mathcal{O},R} = 1$. Thus $\operatorname{div}(M_D) = D - (\sum D) - (\deg D - 1)\mathcal{O}$. For Q_j any point occurring in the 1-fiber O_j define

$$R_j = \frac{M_{Z_j}}{M_{K_j}}, \qquad \boxed{r_j(P) = \frac{R_j(P)}{R_j(Q_j)}}. \tag{13}$$

At removable singularities the quotient is evaluated by its first nonzero local Taylor coefficient. These formulas determine r_j as a rational function in $\mathsf{K}(E)$, without any unspecified constants. The required zero, pole and one fibers of each selected r_j are precisely Z_j, K_j, O_j, with multiplicities, as established in Lemma 4. The formula uses 1,677 maps: 1,621 quadratic and 56 cubic.

4 The exact finite classical-polylogarithmic expression

The main formula is most conveniently stated in terms of *normalized primitives*: each term is defined by an explicit algebraic rational map and vanishes at the initial point of the path. This normalization is essential; it is what prevents an unknown integration constant from surviving the tensor identity. We first define the primitive terms, then give the finite sum.

For any path $q_0 \to P$ avoiding all marked zeros and poles, continue the logarithms and classical polylogarithms analytically along the image under each r_j. All branches below are fixed by continuation from q_0; no principal-branch substitution at the final endpoint is implied. Define

$$L_j(P) = \int_{q_0}^{P} \mathrm{d}\log r_j, \qquad A_j = \log(1 - r_j(q_0)), \tag{14}$$

$$\mathcal{D}_j(P) = \operatorname{Li}_2(r_j(P))\big|_\gamma - \operatorname{Li}_2(r_j(q_0)) + A_j L_j(P), \tag{15}$$

$$\mathcal{T}_j(P) = \operatorname{Li}_3(r_j(P))\big|_\gamma - \operatorname{Li}_3(r_j(q_0)) - \operatorname{Li}_2(r_j(q_0)) L_j(P) + \tfrac{1}{2} A_j L_j(P)^2. \tag{16}$$

The $\operatorname{Li}_2(r_j(q_0))$ and $\operatorname{Li}_3(r_j(q_0))$ constants use the very same initial branches as the continued functions. All the endpoint arguments $r_j(q_0), r_j(q_1)$ are algebraic and different from $0, 1, \infty$. The explicit lists of maps and all coefficients needed to evaluate these arguments appear below.

Appendix A.2 names 20 rational functions $U_0, \ldots, U_{19}$: each is f, g, h, a function r_j, or $1 - r_j$ as specified by its `function` field. Put $\ell_k(P) = \int_{q_0}^{P} \mathrm{d}\log U_k$. For 1,414 triples (a, b, c) with $a \le b \le c$, Appendix A.5 gives coefficients $p_{abc} \in \mathbb{Q}$. Write

$$\mathcal{P}(P) = \sum_{(a,b,c)} p_{abc} \ell_a(P) \ell_b(P) \ell_c(P). \tag{17}$$

Finally, Appendix A.4 lists all 1,677 nonzero integers c_j together with their complete zero, pole, and one fibers, and Appendix A.3 prints the 24 signed integers b_j directly. There are no unprinted coefficients or modular residue conventions in the mathematical formula.

Theorem 1 (Explicit hyperclosure certificate)**.** *With the preceding definitions and the literal finite coefficient tables printed below, the following is an exact identity of multivalued analytic functions with their chosen basepoint branches:*

$$\boxed{B_6(-5) = -30\left\{\sum_j c_j\,\mathcal{T}_j(q_1) + \log f(q_1)\sum_j b_j\,\mathcal{D}_j(q_1) + \mathcal{P}(q_1)\right\}.} \tag{18}$$

Every term is a finite product of logarithms or classical dilogarithms or trilogarithms at algebraic arguments, with algebraic continuation prescribed by γ. Consequently $B_6(-5)$ lies in the hyperclosure ring of Borwein–Chan–Crandall.

5 Proof: an exact tensor certificate and Chen integration

The proof has three independent ingredients. Classical-polylogarithmic primitives integrate the elementary $d\log$ words; the printed coefficient tables satisfy an exact, unquotiented tensor identity; and the prescribed rational maps have the required algebraic fibers, even when a cubic fiber is ramified. Once these statements are established, Chen integration turns the tensor identity into the value formula without an undetermined constant.

Lemma 2 (Polylogarithmic Chen primitives)**.** *If $v = \mathrm{d}\log r$ and $w = \mathrm{d}\log(1-r)$, the functions in (14)–(16), vanishing at q_0, obey*

$$\mathcal{D}_r(P) = -I_{wv}(q_0, P), \qquad \mathcal{T}_r(P) = -I_{wvv}(q_0, P). \tag{19}$$

Proof. For analytically continued polylogarithms, $\mathrm{d}\,\mathrm{Li}_2(r) = -\log(1-r)\,\mathrm{d}\log r$ and $\mathrm{d}\,\mathrm{Li}_3(r) = \mathrm{Li}_2(r)\,\mathrm{d}\log r$. Set $L = \int v$ and $A_0 = \log(1 - r(q_0))$. Then $\mathrm{d}\mathcal{D}_r = -(\log(1-r) - A_0)v = -I_w v$. Further $\mathrm{d}\mathcal{T}_r = (\mathcal{D}_r - A_0 L + A_0 L)v = \mathcal{D}_r v = -I_{wv} v$. All four quantities vanish at q_0, proving both equalities by integration along the chosen path. The calculation remains valid when $r(\gamma)$ crosses a principal branch cut because every derivative is taken along its analytic continuation. □

Lemma 3 (An exact 24-letter tensor certificate)**.** *In the third tensor power of the rational vector space freely spanned by the 24 marked divisor letters, the following equality holds* before taking any shuffle quotient*:*

$$\mathbf{f}\otimes\mathbf{g}\otimes\mathbf{h} = -\sum_j c_j\,\mathbf{w}_j\otimes\mathbf{v}_j\otimes\mathbf{v}_j - \sum_j b_j\,\mathbf{f}\,\text{Ш}\,(\mathbf{w}_j\otimes\mathbf{v}_j) + \sum_{a\le b\le c} p_{abc}\,\mathbf{u}_a\,\text{Ш}\,\mathbf{u}_b\,\text{Ш}\,\mathbf{u}_c. \tag{20}$$

Here $\mathbf{f}, \mathbf{g}, \mathbf{h}, \mathbf{u}_a$ are the integer divisor-order vectors of f, g, h, U_a; $\mathbf{v}_j$ and $\mathbf{w}_j$ are the vectors of r_j and $1 - r_j$. The symbol Ш *denotes the ordinary interleaving/shuffle product, with multiplicities even when letters coincide.*

Proof. The equality is an explicit finite identity over $\mathbb{Q}$. Here is a coordinate-level verification prescription requiring only the data printed in this article. For a divisor vector $\mathbf{a}$ write a_p for its coefficient at the marked point P_p, $0 \le p < 24$. For each ordered triple $(p, q, r) \in \{0, \dots, 23\}^3$, expand the difference between the two sides of (20) as

$$\begin{aligned} 0 = {} & f_p g_q h_r + \sum_j c_j (w_j)_p (v_j)_q (v_j)_r \\ & + \sum_j b_j \left[f_p (w_j)_q (v_j)_r + (w_j)_p f_q (v_j)_r + (w_j)_p (v_j)_q f_r \right] \\ & - \sum_{a\le b\le c} p_{abc} \sum_{\sigma\in S_3} (u_{a_{\sigma(1)}})_p (u_{a_{\sigma(2)}})_q (u_{a_{\sigma(3)}})_r, \end{aligned}$$

where $(a_1, a_2, a_3) = (a, b, c)$ and permutations are counted with multiplicity. The vectors are the signed zero-minus-pole divisors in Appendix A.2 and Appendix A.4, and all c_j, b_j, p_{abc} are literal rational coefficients printed there. Substitution and collection of coefficients yields zero for each of the $24^3 = 13,824$ triples, with exact rational arithmetic; no quotient by shuffles, numerical approximation, or reduction modulo primes is used. This coordinate identity is precisely (20). □

Lemma 4 (Characteristic-zero validity of the map alphabet)**.** *Every r_j occurring in (20) is a function in $\mathsf{K}(E)$ with the specified 0, ∞, and 1 fibers, including multiplicities.*

Proof. Substitution of (10) into (9) gives 24 distinct points on $y^2 = x^3 - 4x$, with coordinates in K. For each of the 1,621 distinct degree-two maps, the three effective disjoint degree-two divisors Z_j, K_j, O_j have the same elliptic group sum. By Abel's theorem they belong to one complete degree-two linear system. By Riemann–Roch, $h^0(D) = 2$ for any degree-two divisor D on an elliptic curve: the linear system is a pencil. The unique ratio M_Z/M_K is nonconstant and degree two, and its value at any point of O_j normalizes the entire 1-fiber to O_j.

For the 56 cubic maps actually used, Abel's condition alone does not establish the prescribed 1-fiber. Form R_j using (12). At each distinct $Q \in O_j$, evaluate the quotient using its first nonzero coefficient in the local coordinate $x - x(Q)$; the resulting values agree and are nonzero. In the seven double-plus-single cases, the first derivative of the normalized map at the double point vanishes. The other 49 cases have three distinct points in O_j. These assertions are identities in the explicitly specified field K, obtained by substitution in the elliptic chord–tangent formulas. Since the fibers have total degree three, no further preimages of 1 exist. Thus each map has the required three fibers over characteristic zero, without recourse to numerical interpolation. □

Proof of Theorem 1. The 24 marked points avoid the physical arc: the f-points have nonreal u-coordinates, the g-points have imaginary u-coordinates, and the two real h-abscissae $2\sqrt{3} \pm \sqrt{10}$ lie outside $[2 + \sqrt{2}, 2\sqrt{2} + \sqrt{6}]$. Hence all r_j and $1 - r_j$ are nonzero and finite on γ, so their logarithmic forms and continued branches are regular along the required integration path.

The valuation-divisor map $\mathsf{K}(E)^\times/\mathsf{K}^\times \to \mathbb{Z}^{24}$ is injective on the group of S-units supported on our marked alphabet. Accordingly, the exact divisor equality (20) implies an equality of tensors of genuine logarithmic one-forms on $E \setminus S$. Apply Chen integration from q_0 to a variable endpoint P. Each pure tensor integrates to its ordered iterated integral; shuffle products integrate to ordinary products of their shorter iterated integrals. By Lemma 2, the two classical-polylogarithmic summands integrate to $\mathcal{T}_j$ and $\log f\, \mathcal{D}_j$. Each one-letter factor integrates to ℓ_k, giving precisely $\mathcal{P}(P)$. Thus

$$I_{fgh}(q_0, P) = \sum_j c_j \mathcal{T}_j(P) + \log f(P) \sum_j b_j \mathcal{D}_j(P) + \mathcal{P}(P)$$

for every endpoint along the physical path. Every term vanishes at $P = q_0$, so there is no undetermined integration constant. At $P = q_1$, multiply by -30 as in (8) to obtain (18).

The endpoint values and all r_j-arguments belong to the number field K adjoined to the algebraic endpoint coordinates. Finally, for algebraic z,

$$\mathrm{Li}_2(z) = z\, {}_3F_2(1, 1, 1; 2, 2; z), \qquad \mathrm{Li}_3(z) = z\, {}_4F_3(1, 1, 1, 1; 2, 2, 2; z),$$

and logarithms are hypergeometric as well. Analytic continuation and inversion formulas allow the arguments to be brought to convergent hypergeometric evaluations when needed. As the hyperclosure ring is closed under finite sums and products, the stated finite expression belongs to it. □

6 Completion of the original $B_6(1)$ problem

We now close the two remaining seeds in Theorem 7.2 of [2]. This section uses only classical dilogarithms, elementary logarithms and the box recurrences; it does not rely on any additional elliptic-period computation. For algebraic $A > 0$, put

$$C_{m,0}(s, A) = \int_{[0,1]^m} (A + x_1^2 + \cdots + x_m^2)^{s/2} \, \mathrm{d}^m x. \tag{21}$$

6.1 The hyperclosed $B_6(-7)$ seed

Define the algebraic and elementary parameters

$$\alpha = \tfrac{1}{2}\arccos(1/4), \quad \beta = \arctan(1/\sqrt{2}), \quad q = \sqrt{10} - 3, \quad w = e^{-i\alpha}, \quad a = \sqrt{6} - \sqrt{5}, \quad b = \sqrt{6} + \sqrt{5}.$$

For $c \in \{\pm ia, \pm ib\}$ and $p \in \{w, -w\}$ define

$$\begin{aligned} \mathscr{F}(c, p; R) &= \log(1 - pR) \log\left(\frac{p(R - c)}{1 - pc}\right) + \mathrm{Li}_2\left(\frac{1 - pR}{1 - pc}\right) - \mathrm{Li}_2\left(\frac{1}{1 - pc}\right), \\ \mathscr{S}(p) &= \mathscr{F}(ia, p; q) - \mathscr{F}(-ia, p; q) - \mathscr{F}(ib, p; q) + \mathscr{F}(-ib, p; q). \end{aligned} \tag{22}$$

All terms here use their principal branches, which are consistent with integration along $[0, q]$ for this choice of parameters.

Proposition 5 (Dilogarithmic evaluation of $B_6(-7)$)**.** *The analytically continued box integral has the finite evaluation*

$$\boxed{B_6(-7) = -\frac{\pi^2}{\sqrt{8}} + \sqrt{8}\,\pi\beta - \frac{8}{\sqrt{2}}\left(\alpha\beta - \frac{1}{2}\Re \sum_{p \in \{w, -w\}} \mathscr{S}(p)\right).} \tag{23}$$

In particular $B_6(-7) \in \mathcal{H}$.

Proof. The starting identity of [2, §7.2] is

$$B_6(-7) = -\frac{\pi^2}{\sqrt{8}} + \sqrt{8}\,\pi\beta - 8I, \qquad I = \int_0^1 \int_0^1 \frac{\mathrm{d}x\,\mathrm{d}y}{(1 + x^2)(1 + y^2)\sqrt{4 + x^2 + y^2}}.$$

Integrate first in y; the substitutions $z = \sqrt{2}x/\sqrt{3 + x^2}$ and $z = 2\sqrt{5}t/(1 + t^2)$ yield

$$I = \frac{1}{\sqrt{2}} \int_0^q \left(\frac{a}{t^2 + a^2} - \frac{b}{t^2 + b^2}\right) (\alpha + \Im \log(1 - w^2 t^2)) \, \mathrm{d}t.$$

The rational factor has integral β on $[0, q]$ and equals $\frac{1}{2i}\left[(t - ia)^{-1} - (t + ia)^{-1} - (t - ib)^{-1} + (t + ib)^{-1}\right]$. Direct differentiation of (22) gives $\frac{\mathrm{d}}{\mathrm{d}R}\mathscr{F}(c, p; R) = \log(1 - pR)/(R - c)$, with $\mathscr{F}(c, p; 0) = 0$. Partial fractions therefore produce (23). Every polylogarithm argument and logarithm argument is algebraic, so the result is hyperclosed. □

6.2 The shifted three-dimensional dilogarithm

The other required input is $D = C_{3,0}(-3,3)$. Elementary integration in one coordinate, then polar coordinates in the remaining square, gives

$$D = \frac{\pi}{4}\log 3 + \int_0^r \left(\frac{h_-}{t^2+h_-^2} + \frac{h_+}{t^2+h_+^2}\right)\log\frac{t^2+\rho^2}{1+\rho^2 t^2}\,\mathrm{d}t, \tag{24}$$

where $\rho = (\sqrt{5}-1)/2$, $h_\pm = \sqrt{5}\pm 2$, $r = \sqrt{6}-\sqrt{5}$. For completeness, the following *finite* dilogarithmic formula eliminates the integral in (24). With $\mathscr{F}$ as in (22) (now allowing purely imaginary c, p), set

$$\widehat{\mathscr{F}}(c,p;r) = \begin{cases} \mathscr{F}(c,p;r) - 2\pi i\log(1/(1-pc)), & c = ih_-,\ p\in\{-i/\rho, -i\rho\},\\ \mathscr{F}(c,p;r), & \text{otherwise.}\end{cases}$$

The displayed correction accounts for the principal-branch jump of $\mathrm{Li}_2(1/(1-pc))$ in exactly the two exceptional cases; otherwise the principal values furnish the continuous primitive along $[0,r]$. Define

$$\mathscr{W}(c) = \sum_{\sigma=\pm1}\left[\widehat{\mathscr{F}}(c,\sigma i/\rho;r) - \widehat{\mathscr{F}}(c,\sigma i\rho;r)\right], \tag{25}$$

$$D = \frac{\pi}{4}\log 3 + 2\log\rho\left(\arctan\frac{r}{h_-} + \arctan\frac{r}{h_+}\right) + \frac{1}{2i}\sum_{h\in\{h_-,h_+\}}\left[\mathscr{W}(ih) - \mathscr{W}(-ih)\right].$$

Indeed, the logarithm in (24) is $2\log\rho + \sum_{\sigma=\pm1}\left[\log(1-\sigma it/\rho) - \log(1-\sigma i\rho t)\right]$, and the rational factor is $\frac{1}{2i}\sum_h\left[(t-ih)^{-1} - (t+ih)^{-1}\right]$. Integrating term by term gives (25). Consequently $D\in\mathcal{H}$; the finite expression evaluates to

$$D = 0.12890150244759601509281247065504226 00\ldots.$$

6.3 Recurrence elimination and the complete conclusion

Theorem 6 (Hyperclosure at every finite integer parameter)**.** *For every integer $s\neq -6$, the meromorphically continued value $B_6(s)$ is hyperclosed. At $s=-6$ the continuation has a simple pole with residue $\pi^3/64$. In particular, the original six-dimensional mean-distance integral satisfies the completely finite expression (5), where its three other non-elementary quantities are given explicitly by (18), (23), and (25).*

Proof. The two recurrences [2, (7.2)–(7.3)] are

$$2sC_{4,0}(s-2,2) = (s+4)C_{4,0}(s,2) - 4C_{3,0}(s,3), \tag{26}$$

$$(s+6)(s+5)B_6(s) = s(s+4)B_6(s-2) + 30C_{4,0}(s,2). \tag{27}$$

At $s=-5$, the second gives $C_{4,0}(-5,2) = -B_6(-7)/6$. Hence the first at $s=-3$ and the second at $s=-3$ yield

$$B_6(-3) = -\tfrac{1}{2}B_6(-5) + 5B_6(-7) + 20D. \tag{28}$$

At $s=-1,1$, the same recurrences give

$$42B_6(1) = \tfrac{3}{8}B_6(-5) - \tfrac{67}{4}B_6(-7) - 67D + 26C_{3,0}(-1,3) + 24C_{3,0}(1,3).$$

For completeness, integrating the divergence of $x(A+|x|^2)^{s/2}$ over the unit m-cube yields

$$(s+m)C_{m,0}(s,A) - sA\,C_{m,0}(s-2,A) = mC_{m-1,0}(s,A+1). \tag{29}$$

Applying it for $m = 3$ and $s = -1, 1$ eliminates the two remaining shifted three-dimensional integrals and produces

$$\boxed{42B_6(1) = \frac{3}{8}B_6(-5) - \frac{67}{4}B_6(-7) - 133D + 66C_{2,0}(-1,4) + 18C_{2,0}(1,4).} \tag{30}$$

Direct polar integration in the square gives, with $L = \log((1+\sqrt{6})/\sqrt{5})$ and $\theta = \arctan\sqrt{2/3}$,

$$C_{2,0}(-1,4) = 2L + 4\theta - \pi, \qquad C_{2,0}(1,4) = \frac{\sqrt{6}}{3} + \frac{13L}{3} + \frac{16\theta}{3} - \frac{4\pi}{3}.$$

Substituting these formulas into (30) proves (5). Theorem 1, Proposition 5, and (25) show that all three exceptional seeds belong to $\mathcal{H}$. Theorem 7.2 of [2] then gives all odd-integer values; its Theorem 7.1 supplies all even-integer values where the meromorphic continuation is finite. The only excluded integer is $s = -6$, as established in [2, §3]. □

7 Independent numerical comparisons

The finite expression may also be evaluated numerically by continuing its logarithmic and polylogarithmic branches along γ. As a comparison away from the box-integral endpoint, the same finite formula at $t = 1/6$ yields

$$I_{fgh} = -0.00277908446967406026792414747668989200817\ldots,$$

in agreement with direct nested Chen-integral quadrature. At $t = 1/3$ the resulting value is

$$\boxed{B_6(-5) = 0.70227312325610135940870498837302070121506914786543235778\ldots}. \tag{31}$$

An independent benchmark from the original six-dimensional integral is

$$B_6(-5) = \frac{\pi^{5/2}}{24}\int_0^\infty \frac{\operatorname{erf}(u)^6}{u^2}\,\mathrm{d}u,$$

whose numerical value is $0.70227312325610135940870498837302070121506914786543235778\ldots$. The endpoint formula involves significant cancellation, including terms of magnitude about 130; numerical agreement by itself is not a proof of the tensor identity.

For the motivating integral, the convergent Laplace representation

$$B_6(1) = \frac{1}{\sqrt{\pi}}\int_0^\infty \frac{1}{u^2}\left[1 - \left(\frac{\sqrt{\pi}\,\operatorname{erf}(u)}{2u}\right)^6\right]\mathrm{d}u = 1.38857408445734784253025407303078881591\ldots$$

provides a separate numerical comparison with (5). Near $u = 0$, the integrand should be evaluated through its Taylor expansion to avoid subtractive cancellation. These comparisons are corroborative; the exact result rests on the algebraic identities, tensor equality, and box recurrences in the preceding sections.

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A Complete literal formula and divisor data

This appendix is part of the formula, not an optional numerical supplement. All three sets of coefficients, all 1,677 used maps, and the entire logarithmic basis are printed explicitly. To evaluate Theorem 1, no external coefficient file is required. The printed map j is the unique rational function defined by equations (11)–(13) with its (Z_j, K_j, O_j) displayed in Table A.4. For each divisor, indices separated by a dot denote points with their given multiplicities (so `05.05` means $2P_5$). The table's d column is the degree, and all coefficients in the last column are literal signed integers.

A.1 Numbering of the twenty-four points

j	family	ϵ	δ	η	j	family	ϵ	δ	η
0	f	-1	-1	-1	12	g	$+1$	-1	-1
1	f	-1	-1	$+1$	13	g	$+1$	-1	$+1$
2	f	-1	$+1$	-1	14	g	$+1$	$+1$	-1
3	f	-1	$+1$	$+1$	15	g	$+1$	$+1$	$+1$
4	f	$+1$	-1	-1	16	h	-1	-1	-1
5	f	$+1$	-1	$+1$	17	h	-1	-1	$+1$
6	f	$+1$	$+1$	-1	18	h	-1	$+1$	-1
7	f	$+1$	$+1$	$+1$	19	h	-1	$+1$	$+1$
8	g	-1	-1	-1	20	h	$+1$	-1	-1
9	g	-1	-1	$+1$	21	h	$+1$	-1	$+1$
10	g	-1	$+1$	-1	22	h	$+1$	$+1$	-1
11	g	-1	$+1$	$+1$	23	h	$+1$	$+1$	$+1$

$$P_j = (x(u_j, Y_j), y(u_j, Y_j)) \tag{32}$$

with (u_j, Y_j) given by equation (10) and the signs above.

A.2 The twenty logarithmic units

The divisor table below follows from the explicit functions in (7) and the normalized map fibers in (13). In each case the function named in the second column determines the divisor shown in the third. The map indices refer to Appendix A.4.

k	U_k	divisor Z/K	k	U_k	divisor Z/K
0	f	`04.05.06.07/00.01.02.03`	10	r_{28}	`04.05/10.11`
1	g	`10.11.14.15/08.09.12.13`	11	r_{30}	`04.05/18.19`
2	h	`16.18.20.22/17.19.21.23`	12	r_{31}	`04.05/22.23`
3	r_0	`05.05/03.03`	13	r_{32}	`04.05/20.21`
4	$1 - r_0$	`00.00/03.03`	14	r_{33}	`04.05/16.17`
5	r_1	`05.05/06.06`	15	r_{1346}	`01.05/11.14`
6	r_{24}	`04.05/02.03`	16	r_{1348}	`01.05/08.13`
7	$1 - r_{24}$	`00.01/02.03`	17	r_{1350}	`01.05/18.22`
8	r_{26}	`04.05/14.15`	18	r_{2666}	`02.05/08.14`
9	r_{27}	`04.05/08.09`	19	r_{2670}	`02.05/18.20`

The divisor column records the zero-minus-pole divisor of each named function; it is not an independent choice of logarithmic units.

A.3 Logarithm–dilogarithm coefficients

j	b_j	j	b_j	j	b_j	j	b_j
60	-1	61	-1	62	+1	63	+1
70	+1	71	+1	72	-1	73	-1
80	-1	81	-1	82	+1	83	+1
90	+1	91	+1	92	-1	93	-1
1380	+2	1382	-2	1390	+2	1392	-2
1400	-2	1402	+2	1410	-2	1412	+2

A.4 All 1,677 trilogarithmic coefficients and rational maps

The entry (Z, K, O) defines the *entire* map by the elliptic chord–tangent Miller formulas; there is no unspecified search-dependent choice.

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
0	2	05.05	03.03	00.00	+34	1	2	05.05	06.06	00.00	−4
2	2	05.05	00.00	03.03	+16	3	2	05.05	06.06	03.03	−16
4	2	05.05	00.00	06.06	−6	5	2	05.05	03.03	06.06	−24
9	2	00.00	06.06	03.03	+28	11	2	00.00	03.03	06.06	+2
24	2	04.05	02.03	00.01	−110	25	2	04.05	06.07	00.01	+164
26	2	04.05	14.15	00.01	−16	27	2	04.05	08.09	00.01	+8
28	2	04.05	10.11	00.01	−6	29	2	04.05	12.13	00.01	+2
30	2	04.05	18.19	00.01	−26	31	2	04.05	22.23	00.01	−26
32	2	04.05	20.21	00.01	−22	33	2	04.05	16.17	00.01	−22
34	2	04.05	00.01	02.03	+264	35	2	04.05	06.07	02.03	−34
36	2	04.05	14.15	02.03	−46	37	2	04.05	08.09	02.03	−26
38	2	04.05	10.11	02.03	−14	39	2	04.05	12.13	02.03	−58
40	2	04.05	18.19	02.03	−82	41	2	04.05	22.23	02.03	−82
42	2	04.05	20.21	02.03	−76	43	2	04.05	16.17	02.03	−76
44	2	04.05	00.01	06.07	+382	45	2	04.05	02.03	06.07	−190
46	2	04.05	14.15	06.07	−42	47	2	04.05	08.09	06.07	−30
48	2	04.05	10.11	06.07	−36	49	2	04.05	12.13	06.07	−24
50	2	04.05	18.19	06.07	−78	51	2	04.05	22.23	06.07	−70
52	2	04.05	20.21	06.07	−56	53	2	04.05	16.17	06.07	−48
54	2	04.05	00.01	14.15	−118	55	2	04.05	02.03	14.15	+138
56	2	04.05	06.07	14.15	−70	57	2	04.05	08.09	14.15	+4
58	2	04.05	10.11	14.15	−58	59	2	04.05	12.13	14.15	+12
60	2	04.05	18.19	14.15	+45	61	2	04.05	22.23	14.15	+53
62	2	04.05	20.21	14.15	+59	63	2	04.05	16.17	14.15	+51
64	2	04.05	00.01	08.09	−136	65	2	04.05	02.03	08.09	+104
66	2	04.05	06.07	08.09	−14	67	2	04.05	14.15	08.09	+4
68	2	04.05	10.11	08.09	−36	69	2	04.05	12.13	08.09	−26
70	2	04.05	18.19	08.09	+61	71	2	04.05	22.23	08.09	+61
72	2	04.05	20.21	08.09	+45	73	2	04.05	16.17	08.09	+45
74	2	04.05	00.01	10.11	−106	75	2	04.05	02.03	10.11	+146
76	2	04.05	06.07	10.11	+6	77	2	04.05	14.15	10.11	−68
78	2	04.05	08.09	10.11	−40	79	2	04.05	12.13	10.11	+22
80	2	04.05	18.19	10.11	+39	81	2	04.05	22.23	10.11	+31
82	2	04.05	20.21	10.11	+43	83	2	04.05	16.17	10.11	+35
84	2	04.05	00.01	12.13	−96	85	2	04.05	02.03	12.13	+108
86	2	04.05	06.07	12.13	−26	87	2	04.05	14.15	12.13	−12
88	2	04.05	08.09	12.13	−40	89	2	04.05	10.11	12.13	+42
90	2	04.05	18.19	12.13	+29	91	2	04.05	22.23	12.13	+29
92	2	04.05	20.21	12.13	+35	93	2	04.05	16.17	12.13	+19
94	2	04.05	00.01	18.19	−22	95	2	04.05	02.03	18.19	−8
96	2	04.05	06.07	18.19	−28	97	2	04.05	14.15	18.19	+44
98	2	04.05	08.09	18.19	+44	99	2	04.05	10.11	18.19	+28
100	2	04.05	12.13	18.19	+8	101	2	04.05	22.23	18.19	−8
102	2	04.05	20.21	18.19	+4	103	2	04.05	16.17	18.19	−20
104	2	04.05	00.01	22.23	−30	105	2	04.05	02.03	22.23	−36
107	2	04.05	14.15	22.23	+60	108	2	04.05	08.09	22.23	+52
109	2	04.05	10.11	22.23	+36	110	2	04.05	12.13	22.23	+24
111	2	04.05	18.19	22.23	−8	112	2	04.05	20.21	22.23	−36

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
113	2	04.05	16.17	22.23	+4	114	2	04.05	00.01	20.21	−80
115	2	04.05	02.03	20.21	+4	116	2	04.05	06.07	20.21	+22
117	2	04.05	14.15	20.21	+76	118	2	04.05	08.09	20.21	+48
119	2	04.05	10.11	20.21	+48	120	2	04.05	12.13	20.21	+16
121	2	04.05	18.19	20.21	+20	122	2	04.05	22.23	20.21	−28
123	2	04.05	16.17	20.21	−12	124	2	04.05	00.01	16.17	−64
126	2	04.05	06.07	16.17	+18	127	2	04.05	14.15	16.17	+52
128	2	04.05	08.09	16.17	+32	129	2	04.05	10.11	16.17	+32
130	2	04.05	12.13	16.17	−8	132	2	04.05	22.23	16.17	+40
133	2	04.05	20.21	16.17	+8	145	2	00.01	06.07	02.03	−198
146	2	00.01	14.15	02.03	−40	147	2	00.01	08.09	02.03	−28
148	2	00.01	10.11	02.03	−46	149	2	00.01	12.13	02.03	−34
150	2	00.01	18.19	02.03	−58	151	2	00.01	22.23	02.03	−66
152	2	00.01	20.21	02.03	−36	153	2	00.01	16.17	02.03	−44
155	2	00.01	02.03	06.07	−80	156	2	00.01	14.15	06.07	−4
157	2	00.01	08.09	06.07	−48	158	2	00.01	10.11	06.07	−36
159	2	00.01	12.13	06.07	−16	160	2	00.01	18.19	06.07	−46
161	2	00.01	22.23	06.07	−46	162	2	00.01	20.21	06.07	−40
163	2	00.01	16.17	06.07	−40	165	2	00.01	02.03	14.15	−14
166	2	00.01	06.07	14.15	+114	167	2	00.01	08.09	14.15	+44
168	2	00.01	10.11	14.15	−10	169	2	00.01	12.13	14.15	−12
170	2	00.01	18.19	14.15	+19	171	2	00.01	22.23	14.15	+19
172	2	00.01	20.21	14.15	+27	173	2	00.01	16.17	14.15	+27
175	2	00.01	02.03	08.09	−2	176	2	00.01	06.07	08.09	+80
177	2	00.01	14.15	08.09	+60	178	2	00.01	10.11	08.09	−12
179	2	00.01	12.13	08.09	+14	180	2	00.01	18.19	08.09	+13
181	2	00.01	22.23	08.09	+13	182	2	00.01	20.21	08.09	+15
183	2	00.01	16.17	08.09	+15	185	2	00.01	02.03	10.11	+12
186	2	00.01	06.07	10.11	+92	187	2	00.01	14.15	10.11	−24
188	2	00.01	08.09	10.11	−8	189	2	00.01	12.13	10.11	+34
190	2	00.01	18.19	10.11	+19	191	2	00.01	22.23	10.11	+19
192	2	00.01	20.21	10.11	+25	193	2	00.01	16.17	10.11	+25
195	2	00.01	02.03	12.13	+24	196	2	00.01	06.07	12.13	+86
197	2	00.01	14.15	12.13	−28	198	2	00.01	08.09	12.13	+12
199	2	00.01	10.11	12.13	+38	200	2	00.01	18.19	12.13	+31
201	2	00.01	22.23	12.13	+31	202	2	00.01	20.21	12.13	+15
203	2	00.01	16.17	12.13	+15	205	2	00.01	02.03	18.19	−26
206	2	00.01	06.07	18.19	+42	207	2	00.01	14.15	18.19	+12
209	2	00.01	10.11	18.19	+20	210	2	00.01	12.13	18.19	+12
211	2	00.01	22.23	18.19	−4	212	2	00.01	20.21	18.19	−24
213	2	00.01	16.17	18.19	−20	215	2	00.01	02.03	22.23	−22
216	2	00.01	06.07	22.23	+30	217	2	00.01	14.15	22.23	+12
219	2	00.01	10.11	22.23	+20	220	2	00.01	12.13	22.23	+12
221	2	00.01	18.19	22.23	−4	222	2	00.01	20.21	22.23	−12
223	2	00.01	16.17	22.23	−16	225	2	00.01	02.03	20.21	+20
226	2	00.01	06.07	20.21	+54	227	2	00.01	14.15	20.21	+12
228	2	00.01	08.09	20.21	−4	229	2	00.01	10.11	20.21	+16
230	2	00.01	12.13	20.21	+4	231	2	00.01	18.19	20.21	+8
232	2	00.01	22.23	20.21	+4	233	2	00.01	16.17	20.21	+8
235	2	00.01	02.03	16.17	+24	236	2	00.01	06.07	16.17	+66
237	2	00.01	14.15	16.17	+12	238	2	00.01	08.09	16.17	−4
239	2	00.01	10.11	16.17	+16	240	2	00.01	12.13	16.17	+4
241	2	00.01	18.19	16.17	−8	242	2	00.01	22.23	16.17	+4
243	2	00.01	20.21	16.17	−4	266	2	02.03	14.15	06.07	+2
267	2	02.03	08.09	06.07	+10	268	2	02.03	10.11	06.07	+12
269	2	02.03	12.13	06.07	+4	270	2	02.03	18.19	06.07	+22
271	2	02.03	22.23	06.07	+22	272	2	02.03	20.21	06.07	+18
273	2	02.03	16.17	06.07	+18	276	2	02.03	06.07	14.15	−66
277	2	02.03	08.09	14.15	−16	278	2	02.03	10.11	14.15	−16
279	2	02.03	12.13	14.15	−58	280	2	02.03	18.19	14.15	−5
281	2	02.03	22.23	14.15	−5	282	2	02.03	20.21	14.15	−11
283	2	02.03	16.17	14.15	−11	286	2	02.03	06.07	08.09	−72
287	2	02.03	14.15	08.09	−16	288	2	02.03	10.11	08.09	−34
289	2	02.03	12.13	08.09	−24	290	2	02.03	18.19	08.09	−5
291	2	02.03	22.23	08.09	−5	292	2	02.03	20.21	08.09	+3
293	2	02.03	16.17	08.09	+3	296	2	02.03	06.07	10.11	−38
297	2	02.03	14.15	10.11	−34	298	2	02.03	08.09	10.11	−62
299	2	02.03	12.13	10.11	+18	300	2	02.03	18.19	10.11	−35

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
301	2	02.03	22.23	10.11	−43	302	2	02.03	20.21	10.11	−23
303	2	02.03	16.17	10.11	−31	306	2	02.03	06.07	12.13	−32
307	2	02.03	14.15	12.13	−58	308	2	02.03	08.09	12.13	−30
309	2	02.03	10.11	12.13	+6	310	2	02.03	18.19	12.13	−33
311	2	02.03	22.23	12.13	−41	312	2	02.03	20.21	12.13	−23
313	2	02.03	16.17	12.13	−31	316	2	02.03	06.07	18.19	+22
317	2	02.03	14.15	18.19	+8	318	2	02.03	08.09	18.19	+16
319	2	02.03	10.11	18.19	−16	320	2	02.03	12.13	18.19	−20
321	2	02.03	22.23	18.19	−44	322	2	02.03	20.21	18.19	+4
323	2	02.03	16.17	18.19	−40	326	2	02.03	06.07	22.23	+14
327	2	02.03	14.15	22.23	+4	328	2	02.03	08.09	22.23	+12
329	2	02.03	10.11	22.23	−20	330	2	02.03	12.13	22.23	−24
331	2	02.03	18.19	22.23	−28	332	2	02.03	20.21	22.23	−24
333	2	02.03	16.17	22.23	+20	336	2	02.03	06.07	20.21	−26
337	2	02.03	14.15	20.21	+24	338	2	02.03	08.09	20.21	+28
339	2	02.03	10.11	20.21	−4	340	2	02.03	12.13	20.21	+4
341	2	02.03	18.19	20.21	−4	342	2	02.03	22.23	20.21	−32
343	2	02.03	16.17	20.21	−40	346	2	02.03	06.07	16.17	−10
347	2	02.03	14.15	16.17	+20	348	2	02.03	08.09	16.17	+24
349	2	02.03	10.11	16.17	−8	351	2	02.03	18.19	16.17	−36
353	2	02.03	20.21	16.17	−36	387	2	06.07	08.09	14.15	+52
388	2	06.07	10.11	14.15	+36	389	2	06.07	12.13	14.15	+22
390	2	06.07	18.19	14.15	−43	391	2	06.07	22.23	14.15	−35
392	2	06.07	20.21	14.15	−27	393	2	06.07	16.17	14.15	−35
397	2	06.07	14.15	08.09	+36	398	2	06.07	10.11	08.09	−2
399	2	06.07	12.13	08.09	+36	400	2	06.07	18.19	08.09	−37
401	2	06.07	22.23	08.09	−37	402	2	06.07	20.21	08.09	−31
403	2	06.07	16.17	08.09	−31	407	2	06.07	14.15	10.11	+30
408	2	06.07	08.09	10.11	−22	409	2	06.07	12.13	10.11	+26
410	2	06.07	18.19	10.11	−31	411	2	06.07	22.23	10.11	−31
412	2	06.07	20.21	10.11	−37	413	2	06.07	16.17	10.11	−37
417	2	06.07	14.15	12.13	+30	418	2	06.07	08.09	12.13	+26
419	2	06.07	10.11	12.13	+30	420	2	06.07	18.19	12.13	−35
421	2	06.07	22.23	12.13	−27	422	2	06.07	20.21	12.13	−19
423	2	06.07	16.17	12.13	−27	427	2	06.07	14.15	18.19	−76
428	2	06.07	08.09	18.19	−56	429	2	06.07	10.11	18.19	−36
430	2	06.07	12.13	18.19	−52	431	2	06.07	22.23	18.19	+24
432	2	06.07	20.21	18.19	+40	433	2	06.07	16.17	18.19	+8
437	2	06.07	14.15	22.23	−56	438	2	06.07	08.09	22.23	−44
439	2	06.07	10.11	22.23	−24	440	2	06.07	12.13	22.23	−32
441	2	06.07	18.19	22.23	+8	442	2	06.07	20.21	22.23	−16
443	2	06.07	16.17	22.23	+32	447	2	06.07	14.15	20.21	−36
448	2	06.07	08.09	20.21	−28	449	2	06.07	10.11	20.21	−24
450	2	06.07	12.13	20.21	−20	451	2	06.07	18.19	20.21	+16
452	2	06.07	22.23	20.21	−16	453	2	06.07	16.17	20.21	+12
457	2	06.07	14.15	16.17	−56	458	2	06.07	08.09	16.17	−40
459	2	06.07	10.11	16.17	−36	460	2	06.07	12.13	16.17	−40
461	2	06.07	18.19	16.17	+4	462	2	06.07	22.23	16.17	+28
463	2	06.07	20.21	16.17	+16	508	2	14.15	10.11	08.09	−4
509	2	14.15	12.13	08.09	+4	518	2	14.15	08.09	10.11	+56
519	2	14.15	12.13	10.11	−56	520	2	14.15	18.19	10.11	+8
521	2	14.15	22.23	10.11	+16	522	2	14.15	20.21	10.11	+8
523	2	14.15	16.17	10.11	+16	528	2	14.15	08.09	12.13	+60
529	2	14.15	10.11	12.13	−60	530	2	14.15	18.19	12.13	+8
531	2	14.15	22.23	12.13	+8	532	2	14.15	20.21	12.13	+8
533	2	14.15	16.17	12.13	+8	538	2	14.15	08.09	18.19	+4
539	2	14.15	10.11	18.19	+8	540	2	14.15	12.13	18.19	+12
543	2	14.15	16.17	18.19	+8	548	2	14.15	08.09	22.23	−20
549	2	14.15	10.11	22.23	+8	550	2	14.15	12.13	22.23	+4
552	2	14.15	20.21	22.23	+24	558	2	14.15	08.09	20.21	−12
559	2	14.15	10.11	20.21	−16	560	2	14.15	12.13	20.21	−12
561	2	14.15	18.19	20.21	−4	562	2	14.15	22.23	20.21	+16
563	2	14.15	16.17	20.21	+8	568	2	14.15	08.09	16.17	−4
569	2	14.15	10.11	16.17	+16	570	2	14.15	12.13	16.17	+12
571	2	14.15	18.19	16.17	+4	572	2	14.15	22.23	16.17	−8
573	2	14.15	20.21	16.17	+4	629	2	08.09	12.13	10.11	−60
630	2	08.09	18.19	10.11	+12	631	2	08.09	22.23	10.11	+12
632	2	08.09	20.21	10.11	+12	633	2	08.09	16.17	10.11	+12

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
639	2	08.09	10.11	12.13	−56	640	2	08.09	18.19	12.13	+4
641	2	08.09	22.23	12.13	+12	642	2	08.09	20.21	12.13	+4
643	2	08.09	16.17	12.13	+12	649	2	08.09	10.11	18.19	−8
650	2	08.09	12.13	18.19	+12	651	2	08.09	22.23	18.19	+16
653	2	08.09	16.17	18.19	+8	660	2	08.09	12.13	22.23	+12
661	2	08.09	18.19	22.23	+16	662	2	08.09	20.21	22.23	+8
669	2	08.09	10.11	20.21	−16	670	2	08.09	12.13	20.21	+4
671	2	08.09	18.19	20.21	−4	673	2	08.09	16.17	20.21	+8
679	2	08.09	10.11	16.17	−8	680	2	08.09	12.13	16.17	+36
681	2	08.09	18.19	16.17	+4	682	2	08.09	22.23	16.17	−8
683	2	08.09	20.21	16.17	+4	750	2	10.11	18.19	12.13	−4
751	2	10.11	22.23	12.13	−4	752	2	10.11	20.21	12.13	−4
753	2	10.11	16.17	12.13	−4	760	2	10.11	12.13	18.19	+4
762	2	10.11	20.21	18.19	+16	763	2	10.11	16.17	18.19	+16
770	2	10.11	12.13	22.23	−12	772	2	10.11	20.21	22.23	+8
773	2	10.11	16.17	22.23	+8	780	2	10.11	12.13	20.21	−12
781	2	10.11	18.19	20.21	+12	782	2	10.11	22.23	20.21	+8
783	2	10.11	16.17	20.21	+8	790	2	10.11	12.13	16.17	+4
791	2	10.11	18.19	16.17	+4	793	2	10.11	20.21	16.17	+4
871	2	12.13	22.23	18.19	+8	873	2	12.13	16.17	18.19	+8
881	2	12.13	18.19	22.23	+8	882	2	12.13	20.21	22.23	+16
883	2	12.13	16.17	22.23	−8	891	2	12.13	18.19	20.21	−4
892	2	12.13	22.23	20.21	+16	901	2	12.13	18.19	16.17	−4
902	2	12.13	22.23	16.17	−16	903	2	12.13	20.21	16.17	−4
992	2	18.19	20.21	22.23	+24	993	2	18.19	16.17	22.23	−24
1002	2	18.19	22.23	20.21	+8	1003	2	18.19	16.17	20.21	−8
1012	2	18.19	22.23	16.17	−16	1013	2	18.19	20.21	16.17	+16
1113	2	22.23	16.17	20.21	+16	1123	2	22.23	20.21	16.17	−8
1344	2	01.05	02.06	00.04	−62	1345	2	01.05	03.07	00.04	+140
1346	2	01.05	11.14	00.04	−14	1347	2	01.05	10.15	00.04	−14
1348	2	01.05	08.13	00.04	−2	1349	2	01.05	09.12	00.04	+6
1350	2	01.05	18.22	00.04	−24	1351	2	01.05	19.23	00.04	−46
1352	2	01.05	17.21	00.04	−8	1353	2	01.05	16.20	00.04	−38
1354	2	01.05	00.04	02.06	−330	1355	2	01.05	03.07	02.06	+126
1356	2	01.05	11.14	02.06	+44	1357	2	01.05	10.15	02.06	+40
1358	2	01.05	08.13	02.06	+52	1359	2	01.05	09.12	02.06	+56
1360	2	01.05	18.22	02.06	+54	1361	2	01.05	19.23	02.06	+60
1362	2	01.05	17.21	02.06	+48	1363	2	01.05	16.20	02.06	+54
1364	2	01.05	00.04	03.07	−170	1365	2	01.05	02.06	03.07	+84
1366	2	01.05	11.14	03.07	+8	1367	2	01.05	10.15	03.07	+12
1368	2	01.05	08.13	03.07	−4	1370	2	01.05	18.22	03.07	+44
1371	2	01.05	19.23	03.07	+56	1372	2	01.05	17.21	03.07	+34
1373	2	01.05	16.20	03.07	+22	1374	2	01.05	00.04	11.14	+70
1375	2	01.05	02.06	11.14	−24	1376	2	01.05	03.07	11.14	−82
1377	2	01.05	10.15	11.14	+16	1378	2	01.05	08.13	11.14	+28
1379	2	01.05	09.12	11.14	−46	1380	2	01.05	18.22	11.14	−10
1381	2	01.05	19.23	11.14	−28	1382	2	01.05	17.21	11.14	−26
1383	2	01.05	16.20	11.14	−26	1384	2	01.05	00.04	10.15	+70
1385	2	01.05	02.06	10.15	−20	1386	2	01.05	03.07	10.15	−86
1387	2	01.05	11.14	10.15	+16	1388	2	01.05	08.13	10.15	−46
1389	2	01.05	09.12	10.15	+28	1390	2	01.05	18.22	10.15	−10
1391	2	01.05	19.23	10.15	−28	1392	2	01.05	17.21	10.15	−26
1393	2	01.05	16.20	10.15	−26	1394	2	01.05	00.04	08.13	+94
1395	2	01.05	02.06	08.13	−16	1396	2	01.05	03.07	08.13	−94
1397	2	01.05	11.14	08.13	+24	1398	2	01.05	10.15	08.13	−30
1399	2	01.05	09.12	08.13	−12	1400	2	01.05	18.22	08.13	−22
1401	2	01.05	19.23	08.13	−22	1402	2	01.05	17.21	08.13	−14
1403	2	01.05	16.20	08.13	−16	1404	2	01.05	00.04	09.12	+102
1405	2	01.05	02.06	09.12	−4	1406	2	01.05	03.07	09.12	−82
1407	2	01.05	11.14	09.12	−42	1408	2	01.05	10.15	09.12	+12
1409	2	01.05	08.13	09.12	−24	1410	2	01.05	18.22	09.12	−22
1411	2	01.05	19.23	09.12	−22	1412	2	01.05	17.21	09.12	−14
1413	2	01.05	16.20	09.12	−16	1414	2	01.05	00.04	18.22	−2
1415	2	01.05	02.06	18.22	+16	1416	2	01.05	03.07	18.22	−38
1417	2	01.05	11.14	18.22	−12	1418	2	01.05	10.15	18.22	−12
1419	2	01.05	08.13	18.22	−4	1420	2	01.05	09.12	18.22	−12
1422	2	01.05	17.21	18.22	+8	1423	2	01.05	16.20	18.22	+28
1424	2	01.05	00.04	19.23	−8	1425	2	01.05	02.06	19.23	−26

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
1426	2	01.05	03.07	19.23	−8	1427	2	01.05	11.14	19.23	−12
1428	2	01.05	10.15	19.23	−12	1429	2	01.05	08.13	19.23	−4
1430	2	01.05	09.12	19.23	−4	1432	2	01.05	17.21	19.23	+12
1433	2	01.05	16.20	19.23	+24	1434	2	01.05	00.04	17.21	+68
1435	2	01.05	02.06	17.21	−30	1436	2	01.05	03.07	17.21	−58
1437	2	01.05	11.14	17.21	−16	1438	2	01.05	10.15	17.21	−12
1439	2	01.05	08.13	17.21	+8	1440	2	01.05	09.12	17.21	+4
1441	2	01.05	18.22	17.21	−4	1442	2	01.05	19.23	17.21	−8
1443	2	01.05	16.20	17.21	−12	1444	2	01.05	00.04	16.20	+38
1445	2	01.05	02.06	16.20	−8	1446	2	01.05	03.07	16.20	−68
1447	2	01.05	11.14	16.20	−20	1448	2	01.05	10.15	16.20	−24
1449	2	01.05	08.13	16.20	−4	1450	2	01.05	09.12	16.20	−8
1451	2	01.05	18.22	16.20	−4	1452	2	01.05	19.23	16.20	+8
1453	2	01.05	17.21	16.20	−8	1465	2	00.04	03.07	02.06	−14
1466	2	00.04	11.14	02.06	+74	1467	2	00.04	10.15	02.06	+70
1468	2	00.04	08.13	02.06	+62	1469	2	00.04	09.12	02.06	+58
1470	2	00.04	18.22	02.06	+92	1471	2	00.04	19.23	02.06	+80
1472	2	00.04	17.21	02.06	+58	1473	2	00.04	16.20	02.06	+70
1475	2	00.04	02.06	03.07	+146	1476	2	00.04	11.14	03.07	+6
1477	2	00.04	10.15	03.07	+10	1478	2	00.04	08.13	03.07	+22
1479	2	00.04	09.12	03.07	+18	1480	2	00.04	18.22	03.07	+74
1481	2	00.04	19.23	03.07	+68	1482	2	00.04	17.21	03.07	+68
1483	2	00.04	16.20	03.07	+62	1485	2	00.04	02.06	11.14	−64
1486	2	00.04	03.07	11.14	−38	1487	2	00.04	10.15	11.14	+64
1488	2	00.04	08.13	11.14	−44	1489	2	00.04	09.12	11.14	+30
1490	2	00.04	18.22	11.14	−58	1491	2	00.04	19.23	11.14	−30
1492	2	00.04	17.21	11.14	−48	1493	2	00.04	16.20	11.14	−50
1495	2	00.04	02.06	10.15	−92	1496	2	00.04	03.07	10.15	−10
1497	2	00.04	11.14	10.15	+64	1498	2	00.04	08.13	10.15	+6
1499	2	00.04	09.12	10.15	−20	1500	2	00.04	18.22	10.15	−38
1501	2	00.04	19.23	10.15	−50	1502	2	00.04	17.21	10.15	−52
1503	2	00.04	16.20	10.15	−46	1505	2	00.04	02.06	08.13	−96
1506	2	00.04	03.07	08.13	−10	1507	2	00.04	11.14	08.13	−48
1508	2	00.04	10.15	08.13	+22	1509	2	00.04	09.12	08.13	+36
1510	2	00.04	18.22	08.13	−34	1511	2	00.04	19.23	08.13	−20
1512	2	00.04	17.21	08.13	−8	1513	2	00.04	16.20	08.13	−28
1515	2	00.04	02.06	09.12	−68	1516	2	00.04	03.07	09.12	+18
1517	2	00.04	11.14	09.12	+34	1518	2	00.04	10.15	09.12	−36
1519	2	00.04	08.13	09.12	+24	1520	2	00.04	18.22	09.12	−70
1521	2	00.04	19.23	09.12	−64	1522	2	00.04	17.21	09.12	−68
1523	2	00.04	16.20	09.12	−48	1525	2	00.04	02.06	18.22	−6
1526	2	00.04	03.07	18.22	+56	1527	2	00.04	11.14	18.22	−64
1528	2	00.04	10.15	18.22	−44	1529	2	00.04	08.13	18.22	−32
1530	2	00.04	09.12	18.22	−44	1532	2	00.04	17.21	18.22	−8
1533	2	00.04	16.20	18.22	+32	1535	2	00.04	02.06	19.23	+26
1536	2	00.04	03.07	19.23	+20	1537	2	00.04	11.14	19.23	−28
1538	2	00.04	10.15	19.23	−48	1539	2	00.04	08.13	19.23	−12
1540	2	00.04	09.12	19.23	−40	1542	2	00.04	17.21	19.23	+48
1543	2	00.04	16.20	19.23	−24	1545	2	00.04	02.06	17.21	−22
1546	2	00.04	03.07	17.21	−14	1547	2	00.04	11.14	17.21	−52
1548	2	00.04	10.15	17.21	−60	1549	2	00.04	08.13	17.21	−4
1550	2	00.04	09.12	17.21	−44	1551	2	00.04	18.22	17.21	−20
1552	2	00.04	19.23	17.21	+20	1553	2	00.04	16.20	17.21	−4
1555	2	00.04	02.06	16.20	−38	1556	2	00.04	03.07	16.20	+22
1557	2	00.04	11.14	16.20	−44	1558	2	00.04	10.15	16.20	−36
1559	2	00.04	08.13	16.20	−20	1560	2	00.04	09.12	16.20	−20
1561	2	00.04	18.22	16.20	+8	1562	2	00.04	19.23	16.20	−40
1563	2	00.04	17.21	16.20	−16	1586	2	02.06	11.14	03.07	−52
1587	2	02.06	10.15	03.07	−44	1588	2	02.06	08.13	03.07	−48
1589	2	02.06	09.12	03.07	−48	1590	2	02.06	18.22	03.07	−28
1591	2	02.06	19.23	03.07	+2	1592	2	02.06	17.21	03.07	−4
1593	2	02.06	16.20	03.07	−46	1596	2	02.06	03.07	11.14	−4
1597	2	02.06	10.15	11.14	−48	1598	2	02.06	08.13	11.14	+20
1599	2	02.06	09.12	11.14	+32	1600	2	02.06	18.22	11.14	+28
1601	2	02.06	19.23	11.14	+44	1602	2	02.06	17.21	11.14	+24
1603	2	02.06	16.20	11.14	+42	1607	2	02.06	11.14	10.15	−48
1608	2	02.06	08.13	10.15	+8	1609	2	02.06	09.12	10.15	+20
1610	2	02.06	18.22	10.15	+56	1611	2	02.06	19.23	10.15	+56

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
1612	2	02.06	17.21	10.15	+52	1613	2	02.06	16.20	10.15	+54
1616	2	02.06	03.07	08.13	−4	1617	2	02.06	11.14	08.13	+32
1619	2	02.06	09.12	08.13	−36	1620	2	02.06	18.22	08.13	+62
1621	2	02.06	19.23	08.13	+48	1622	2	02.06	17.21	08.13	+46
1623	2	02.06	16.20	08.13	+46	1627	2	02.06	11.14	09.12	+12
1628	2	02.06	10.15	09.12	+20	1629	2	02.06	08.13	09.12	−48
1630	2	02.06	18.22	09.12	+34	1631	2	02.06	19.23	09.12	+36
1632	2	02.06	17.21	09.12	+18	1633	2	02.06	16.20	09.12	+34
1636	2	02.06	03.07	18.22	−36	1637	2	02.06	11.14	18.22	+28
1638	2	02.06	10.15	18.22	+40	1639	2	02.06	08.13	18.22	+60
1640	2	02.06	09.12	18.22	+40	1641	2	02.06	19.23	18.22	+8
1642	2	02.06	17.21	18.22	+8	1643	2	02.06	16.20	18.22	−4
1647	2	02.06	11.14	19.23	+56	1648	2	02.06	10.15	19.23	+60
1649	2	02.06	08.13	19.23	+48	1650	2	02.06	09.12	19.23	+44
1652	2	02.06	17.21	19.23	−12	1653	2	02.06	16.20	19.23	−8
1656	2	02.06	03.07	17.21	+20	1657	2	02.06	11.14	17.21	+28
1658	2	02.06	10.15	17.21	+44	1659	2	02.06	08.13	17.21	+36
1660	2	02.06	09.12	17.21	+20	1661	2	02.06	18.22	17.21	+12
1663	2	02.06	16.20	17.21	+12	1667	2	02.06	11.14	16.20	+28
1668	2	02.06	10.15	16.20	+28	1669	2	02.06	08.13	16.20	+20
1670	2	02.06	09.12	16.20	+12	1671	2	02.06	18.22	16.20	+12
1672	2	02.06	19.23	16.20	−8	1673	2	02.06	17.21	16.20	+8
1708	2	03.07	08.13	11.14	−24	1709	2	03.07	09.12	11.14	−12
1710	2	03.07	18.22	11.14	+22	1711	2	03.07	19.23	11.14	+32
1712	2	03.07	17.21	11.14	+36	1713	2	03.07	16.20	11.14	+8
1718	2	03.07	08.13	10.15	−12	1720	2	03.07	18.22	10.15	+14
1722	2	03.07	17.21	10.15	+4	1727	2	03.07	11.14	08.13	−12
1728	2	03.07	10.15	08.13	−20	1729	2	03.07	09.12	08.13	+12
1730	2	03.07	18.22	08.13	+20	1731	2	03.07	19.23	08.13	+32
1732	2	03.07	17.21	08.13	+22	1733	2	03.07	16.20	08.13	+8
1737	2	03.07	11.14	09.12	−32	1740	2	03.07	18.22	09.12	+12
1742	2	03.07	17.21	09.12	−10	1747	2	03.07	11.14	18.22	+24
1748	2	03.07	10.15	18.22	+32	1749	2	03.07	08.13	18.22	+16
1750	2	03.07	09.12	18.22	+16	1751	2	03.07	19.23	18.22	−8
1752	2	03.07	17.21	18.22	−16	1757	2	03.07	11.14	19.23	+24
1759	2	03.07	08.13	19.23	+24	1762	2	03.07	17.21	19.23	+8
1767	2	03.07	11.14	17.21	+16	1768	2	03.07	10.15	17.21	−4
1769	2	03.07	08.13	17.21	+16	1770	2	03.07	09.12	17.21	−12
1771	2	03.07	18.22	17.21	−12	1772	2	03.07	19.23	17.21	+28
1773	2	03.07	16.20	17.21	+4	1777	2	03.07	11.14	16.20	−4
1779	2	03.07	08.13	16.20	−4	1781	2	03.07	18.22	16.20	+8
1840	2	11.14	18.22	08.13	−12	1841	2	11.14	19.23	08.13	+4
1842	2	11.14	17.21	08.13	−12	1843	2	11.14	16.20	08.13	+4
1850	2	11.14	18.22	09.12	+4	1851	2	11.14	19.23	09.12	+12
1852	2	11.14	17.21	09.12	+12	1853	2	11.14	16.20	09.12	+4
1858	2	11.14	10.15	18.22	+16	1859	2	11.14	08.13	18.22	−4
1860	2	11.14	09.12	18.22	+12	1863	2	11.14	16.20	18.22	−16
1869	2	11.14	08.13	19.23	−4	1870	2	11.14	09.12	19.23	+12
1872	2	11.14	17.21	19.23	−16	1873	2	11.14	16.20	19.23	−8
1878	2	11.14	10.15	17.21	+16	1879	2	11.14	08.13	17.21	−4
1880	2	11.14	09.12	17.21	+20	1881	2	11.14	18.22	17.21	−4
1882	2	11.14	19.23	17.21	−8	1883	2	11.14	16.20	17.21	−4
1889	2	11.14	08.13	16.20	−4	1890	2	11.14	09.12	16.20	+20
1891	2	11.14	18.22	16.20	−12	1892	2	11.14	19.23	16.20	+8
1893	2	11.14	17.21	16.20	+8	1951	2	10.15	19.23	08.13	−8
1952	2	10.15	17.21	08.13	−8	1960	2	10.15	18.22	09.12	+16
1962	2	10.15	17.21	09.12	+16	1970	2	10.15	09.12	18.22	+16
1972	2	10.15	17.21	18.22	−8	1973	2	10.15	16.20	18.22	−16
1982	2	10.15	17.21	19.23	−16	1990	2	10.15	09.12	17.21	+16
1991	2	10.15	18.22	17.21	+4	1992	2	10.15	19.23	17.21	−16
1993	2	10.15	16.20	17.21	+4	1999	2	10.15	08.13	16.20	+16
2001	2	10.15	18.22	16.20	−4	2070	2	08.13	18.22	09.12	+12
2071	2	08.13	19.23	09.12	+12	2072	2	08.13	17.21	09.12	+12
2073	2	08.13	16.20	09.12	+12	2081	2	08.13	19.23	18.22	+8
2082	2	08.13	17.21	18.22	+8	2083	2	08.13	16.20	18.22	−8
2101	2	08.13	18.22	17.21	+4	2103	2	08.13	16.20	17.21	+4
2111	2	08.13	18.22	16.20	+4	2112	2	08.13	19.23	16.20	+16
2113	2	08.13	17.21	16.20	+8	2191	2	09.12	19.23	18.22	−8

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
2192	2	09.12	17.21	18.22	-8	2193	2	09.12	16.20	18.22	-8
2202	2	09.12	17.21	19.23	-16	2211	2	09.12	18.22	17.21	+4
2212	2	09.12	19.23	17.21	-8	2213	2	09.12	16.20	17.21	-4
2221	2	09.12	18.22	16.20	-4	2664	2	02.05	00.07	03.04	-116
2665	2	02.05	01.06	03.04	-8	2666	2	02.05	08.14	03.04	+14
2667	2	02.05	09.15	03.04	+14	2668	2	02.05	10.12	03.04	+18
2669	2	02.05	11.13	03.04	+26	2670	2	02.05	18.20	03.04	+56
2671	2	02.05	19.21	03.04	+24	2672	2	02.05	16.22	03.04	+44
2673	2	02.05	17.23	03.04	+36	2674	2	02.05	03.04	00.07	-60
2675	2	02.05	01.06	00.07	-46	2676	2	02.05	08.14	00.07	+34
2677	2	02.05	09.15	00.07	+30	2678	2	02.05	10.12	00.07	+24
2679	2	02.05	11.13	00.07	+28	2680	2	02.05	18.20	00.07	+34
2681	2	02.05	19.21	00.07	+22	2682	2	02.05	16.22	00.07	+26
2683	2	02.05	17.23	00.07	+14	2684	2	02.05	03.04	01.06	+30
2685	2	02.05	00.07	01.06	-52	2686	2	02.05	08.14	01.06	-8
2687	2	02.05	09.15	01.06	-4	2688	2	02.05	10.12	01.06	-18
2689	2	02.05	11.13	01.06	-14	2690	2	02.05	18.20	01.06	+26
2691	2	02.05	19.21	01.06	+6	2692	2	02.05	16.22	01.06	+22
2693	2	02.05	17.23	01.06	+10	2694	2	02.05	03.04	08.14	-32
2695	2	02.05	00.07	08.14	+108	2696	2	02.05	01.06	08.14	+52
2697	2	02.05	09.15	08.14	+60	2698	2	02.05	10.12	08.14	+50
2699	2	02.05	11.13	08.14	+24	2700	2	02.05	18.20	08.14	-62
2701	2	02.05	19.21	08.14	-50	2702	2	02.05	16.22	08.14	-58
2703	2	02.05	17.23	08.14	-54	2704	2	02.05	03.04	09.15	-36
2705	2	02.05	00.07	09.15	+76	2706	2	02.05	01.06	09.15	+44
2707	2	02.05	08.14	09.15	+72	2708	2	02.05	10.12	09.15	+36
2709	2	02.05	11.13	09.15	+38	2710	2	02.05	18.20	09.15	-38
2711	2	02.05	19.21	09.15	-34	2712	2	02.05	16.22	09.15	-42
2713	2	02.05	17.23	09.15	-30	2714	2	02.05	03.04	10.12	-6
2715	2	02.05	00.07	10.12	+66	2716	2	02.05	01.06	10.12	+8
2717	2	02.05	08.14	10.12	+70	2718	2	02.05	09.15	10.12	+42
2719	2	02.05	11.13	10.12	+32	2720	2	02.05	18.20	10.12	-42
2721	2	02.05	19.21	10.12	-30	2722	2	02.05	16.22	10.12	-30
2723	2	02.05	17.23	10.12	-26	2724	2	02.05	03.04	11.13	-2
2725	2	02.05	00.07	11.13	+42	2726	2	02.05	01.06	11.13	+16
2727	2	02.05	08.14	11.13	+42	2728	2	02.05	09.15	11.13	+46
2729	2	02.05	10.12	11.13	+32	2730	2	02.05	18.20	11.13	-18
2731	2	02.05	19.21	11.13	-14	2732	2	02.05	16.22	11.13	-14
2733	2	02.05	17.23	11.13	-2	2734	2	02.05	03.04	18.20	+70
2735	2	02.05	00.07	18.20	+40	2736	2	02.05	01.06	18.20	-6
2737	2	02.05	08.14	18.20	-52	2738	2	02.05	09.15	18.20	-40
2739	2	02.05	10.12	18.20	-28	2740	2	02.05	11.13	18.20	-24
2741	2	02.05	19.21	18.20	+12	2742	2	02.05	16.22	18.20	+60
2743	2	02.05	17.23	18.20	+16	2744	2	02.05	03.04	19.21	+34
2745	2	02.05	00.07	19.21	+16	2746	2	02.05	01.06	19.21	+26
2747	2	02.05	08.14	19.21	-68	2748	2	02.05	09.15	19.21	-56
2749	2	02.05	10.12	19.21	-40	2750	2	02.05	11.13	19.21	-28
2751	2	02.05	18.20	19.21	+8	2752	2	02.05	16.22	19.21	+12
2753	2	02.05	17.23	19.21	+56	2754	2	02.05	03.04	16.22	+94
2755	2	02.05	00.07	16.22	+32	2756	2	02.05	01.06	16.22	-10
2757	2	02.05	08.14	16.22	-48	2758	2	02.05	09.15	16.22	-32
2759	2	02.05	10.12	16.22	-24	2760	2	02.05	11.13	16.22	-16
2761	2	02.05	18.20	16.22	+40	2762	2	02.05	19.21	16.22	+8
2763	2	02.05	17.23	16.22	-8	2764	2	02.05	03.04	17.23	+58
2766	2	02.05	01.06	17.23	+22	2767	2	02.05	08.14	17.23	-64
2768	2	02.05	09.15	17.23	-56	2769	2	02.05	10.12	17.23	-36
2770	2	02.05	11.13	17.23	-28	2771	2	02.05	18.20	17.23	+8
2772	2	02.05	19.21	17.23	+56	2785	2	03.04	01.06	00.07	-26
2786	2	03.04	08.14	00.07	+6	2787	2	03.04	09.15	00.07	+2
2788	2	03.04	10.12	00.07	+12	2789	2	03.04	11.13	00.07	+8
2790	2	03.04	18.20	00.07	-14	2791	2	03.04	19.21	00.07	-26
2792	2	03.04	16.22	00.07	-10	2793	2	03.04	17.23	00.07	-30
2795	2	03.04	00.07	01.06	+52	2796	2	03.04	08.14	01.06	-32
2797	2	03.04	09.15	01.06	-28	2798	2	03.04	10.12	01.06	-34
2799	2	03.04	11.13	01.06	-38	2800	2	03.04	18.20	01.06	-14
2801	2	03.04	19.21	01.06	-2	2802	2	03.04	16.22	01.06	-22
2803	2	03.04	17.23	01.06	-10	2805	2	03.04	00.07	08.14	+38
2806	2	03.04	01.06	08.14	+46	2807	2	03.04	09.15	08.14	+12

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
2809	2	03.04	11.13	08.14	+70	2810	2	03.04	18.20	08.14	−2
2811	2	03.04	19.21	08.14	−14	2812	2	03.04	16.22	08.14	−2
2813	2	03.04	17.23	08.14	−14	2815	2	03.04	00.07	09.15	+6
2816	2	03.04	01.06	09.15	+38	2817	2	03.04	08.14	09.15	+24
2818	2	03.04	10.12	09.15	+58	2819	2	03.04	11.13	09.15	+12
2820	2	03.04	18.20	09.15	+6	2821	2	03.04	19.21	09.15	+18
2822	2	03.04	16.22	09.15	+6	2823	2	03.04	17.23	09.15	+18
2825	2	03.04	00.07	10.12	−18	2826	2	03.04	01.06	10.12	+8
2827	2	03.04	08.14	10.12	+20	2828	2	03.04	09.15	10.12	+64
2829	2	03.04	11.13	10.12	−32	2830	2	03.04	18.20	10.12	+22
2831	2	03.04	19.21	10.12	+34	2832	2	03.04	16.22	10.12	+30
2833	2	03.04	17.23	10.12	+42	2835	2	03.04	00.07	11.13	+6
2837	2	03.04	08.14	11.13	+88	2838	2	03.04	09.15	11.13	+20
2839	2	03.04	10.12	11.13	−32	2840	2	03.04	18.20	11.13	+14
2841	2	03.04	19.21	11.13	+2	2842	2	03.04	16.22	11.13	+22
2843	2	03.04	17.23	11.13	+10	2845	2	03.04	00.07	18.20	−34
2846	2	03.04	01.06	18.20	+28	2847	2	03.04	08.14	18.20	−28
2848	2	03.04	09.15	18.20	−32	2850	2	03.04	11.13	18.20	+4
2851	2	03.04	19.21	18.20	+20	2853	2	03.04	17.23	18.20	+44
2855	2	03.04	00.07	19.21	−18	2856	2	03.04	01.06	19.21	+4
2857	2	03.04	08.14	19.21	−36	2858	2	03.04	09.15	19.21	−8
2859	2	03.04	10.12	19.21	+12	2860	2	03.04	11.13	19.21	−8
2861	2	03.04	18.20	19.21	+16	2862	2	03.04	16.22	19.21	+40
2863	2	03.04	17.23	19.21	−4	2865	2	03.04	00.07	16.22	−34
2866	2	03.04	01.06	16.22	+8	2867	2	03.04	08.14	16.22	−28
2868	2	03.04	09.15	16.22	−28	2870	2	03.04	11.13	16.22	+8
2871	2	03.04	18.20	16.22	−4	2872	2	03.04	19.21	16.22	+36
2873	2	03.04	17.23	16.22	+8	2875	2	03.04	00.07	17.23	−26
2877	2	03.04	08.14	17.23	−28	2878	2	03.04	09.15	17.23	−4
2879	2	03.04	10.12	17.23	+20	2880	2	03.04	11.13	17.23	−4
2881	2	03.04	18.20	17.23	+36	2882	2	03.04	19.21	17.23	−20
2906	2	00.07	08.14	01.06	−36	2907	2	00.07	09.15	01.06	−28
2908	2	00.07	10.12	01.06	−24	2909	2	00.07	11.13	01.06	−24
2910	2	00.07	18.20	01.06	−16	2911	2	00.07	19.21	01.06	−8
2912	2	00.07	16.22	01.06	−28	2913	2	00.07	17.23	01.06	+4
2917	2	00.07	09.15	08.14	−60	2918	2	00.07	10.12	08.14	+6
2919	2	00.07	11.13	08.14	+8	2921	2	00.07	19.21	08.14	−16
2922	2	00.07	16.22	08.14	+4	2923	2	00.07	17.23	08.14	−20
2927	2	00.07	08.14	09.15	−48	2928	2	00.07	10.12	09.15	−4
2929	2	00.07	11.13	09.15	−6	2930	2	00.07	18.20	09.15	+32
2931	2	00.07	19.21	09.15	+32	2932	2	00.07	16.22	09.15	+28
2933	2	00.07	17.23	09.15	+36	2937	2	00.07	08.14	10.12	+18
2938	2	00.07	09.15	10.12	−2	2940	2	00.07	18.20	10.12	+4
2941	2	00.07	19.21	10.12	+12	2942	2	00.07	16.22	10.12	+8
2943	2	00.07	17.23	10.12	+8	2947	2	00.07	08.14	11.13	+22
2948	2	00.07	09.15	11.13	−6	2950	2	00.07	18.20	11.13	+20
2951	2	00.07	19.21	11.13	−4	2952	2	00.07	16.22	11.13	+16
2957	2	00.07	08.14	18.20	+24	2958	2	00.07	09.15	18.20	+28
2959	2	00.07	10.12	18.20	+8	2960	2	00.07	11.13	18.20	+20
2961	2	00.07	19.21	18.20	+4	2963	2	00.07	17.23	18.20	+12
2967	2	00.07	08.14	19.21	+12	2968	2	00.07	09.15	19.21	+48
2969	2	00.07	10.12	19.21	+16	2970	2	00.07	11.13	19.21	+4
2971	2	00.07	18.20	19.21	−8	2972	2	00.07	16.22	19.21	+8
2973	2	00.07	17.23	19.21	−20	2977	2	00.07	08.14	16.22	+24
2978	2	00.07	09.15	16.22	+32	2979	2	00.07	10.12	16.22	+8
2980	2	00.07	11.13	16.22	+24	2981	2	00.07	18.20	16.22	−4
2982	2	00.07	19.21	16.22	+12	2987	2	00.07	08.14	17.23	+12
2988	2	00.07	09.15	17.23	+44	2989	2	00.07	10.12	17.23	+16
2991	2	00.07	18.20	17.23	+12	2992	2	00.07	19.21	17.23	−4
3027	2	01.06	09.15	08.14	−12	3028	2	01.06	10.12	08.14	−12
3029	2	01.06	11.13	08.14	−14	3038	2	01.06	10.12	09.15	−2
3067	2	01.06	08.14	18.20	+8	3068	2	01.06	09.15	18.20	+12
3069	2	01.06	10.12	18.20	+4	3071	2	01.06	19.21	18.20	−4
3072	2	01.06	16.22	18.20	−12	3077	2	01.06	08.14	19.21	−4
3079	2	01.06	10.12	19.21	−4	3082	2	01.06	16.22	19.21	−4
3087	2	01.06	08.14	16.22	+4	3088	2	01.06	09.15	16.22	+12
3150	2	08.14	18.20	09.15	−12	3151	2	08.14	19.21	09.15	−12
3152	2	08.14	16.22	09.15	−12	3153	2	08.14	17.23	09.15	−12

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
3160	2	08.14	18.20	10.12	−12	3161	2	08.14	19.21	10.12	−20
3162	2	08.14	16.22	10.12	−12	3163	2	08.14	17.23	10.12	−20
3170	2	08.14	18.20	11.13	−8	3171	2	08.14	19.21	11.13	−16
3172	2	08.14	16.22	11.13	−24	3173	2	08.14	17.23	11.13	−16
3180	2	08.14	11.13	18.20	+16	3181	2	08.14	19.21	18.20	−12
3182	2	08.14	16.22	18.20	−12	3183	2	08.14	17.23	18.20	−24
3190	2	08.14	11.13	19.21	+16	3191	2	08.14	18.20	19.21	−8
3192	2	08.14	16.22	19.21	−4	3193	2	08.14	17.23	19.21	−8
3199	2	08.14	10.12	16.22	−8	3201	2	08.14	18.20	16.22	−8
3202	2	08.14	19.21	16.22	−16	3203	2	08.14	17.23	16.22	−8
3209	2	08.14	10.12	17.23	−8	3211	2	08.14	18.20	17.23	−16
3212	2	08.14	19.21	17.23	−8	3270	2	09.15	18.20	10.12	−4
3271	2	09.15	19.21	10.12	+4	3272	2	09.15	16.22	10.12	−4
3273	2	09.15	17.23	10.12	−12	3280	2	09.15	18.20	11.13	−8
3282	2	09.15	16.22	11.13	−8	3289	2	09.15	10.12	18.20	+8
3291	2	09.15	19.21	18.20	−4	3292	2	09.15	16.22	18.20	−4
3299	2	09.15	10.12	19.21	+24	3302	2	09.15	16.22	19.21	−12
3310	2	09.15	11.13	16.22	−16	3312	2	09.15	19.21	16.22	−8
3313	2	09.15	17.23	16.22	+8	3321	2	09.15	18.20	17.23	−8
3401	2	10.12	19.21	18.20	−4	3402	2	10.12	16.22	18.20	−4
3403	2	10.12	17.23	18.20	−16	3411	2	10.12	18.20	19.21	−8
3412	2	10.12	16.22	19.21	−12	3431	2	10.12	18.20	17.23	−16
3511	2	11.13	19.21	18.20	−12	3512	2	11.13	16.22	18.20	−4
3513	2	11.13	17.23	18.20	−16	3522	2	11.13	16.22	19.21	−12
3532	2	11.13	19.21	16.22	−16	3984	2	05.07	00.02	04.06	+20
3985	2	05.07	01.03	04.06	+30	3986	2	05.07	12.14	04.06	−20
3987	2	05.07	13.15	04.06	−20	3988	2	05.07	08.10	04.06	−16
3989	2	05.07	09.11	04.06	−24	3990	2	05.07	17.18	04.06	−18
3991	2	05.07	16.19	04.06	+4	3992	2	05.07	21.22	04.06	−6
3993	2	05.07	20.23	04.06	−8	3994	2	05.07	04.06	00.02	−56
3995	2	05.07	01.03	00.02	+30	3996	2	05.07	12.14	00.02	+2
3997	2	05.07	13.15	00.02	+6	3998	2	05.07	08.10	00.02	−26
3999	2	05.07	09.11	00.02	−30	4000	2	05.07	17.18	00.02	+28
4001	2	05.07	16.19	00.02	+22	4002	2	05.07	21.22	00.02	+28
4003	2	05.07	20.23	00.02	+22	4004	2	05.07	04.06	01.03	−46
4005	2	05.07	00.02	01.03	+10	4006	2	05.07	12.14	01.03	+12
4007	2	05.07	13.15	01.03	+8	4008	2	05.07	08.10	01.03	+20
4009	2	05.07	09.11	01.03	+16	4010	2	05.07	17.18	01.03	+14
4011	2	05.07	16.19	01.03	+2	4012	2	05.07	21.22	01.03	+18
4013	2	05.07	20.23	01.03	−2	4014	2	05.07	04.06	12.14	+24
4015	2	05.07	00.02	12.14	−34	4016	2	05.07	01.03	12.14	−36
4017	2	05.07	13.15	12.14	−28	4018	2	05.07	08.10	12.14	−40
4019	2	05.07	09.11	12.14	+10	4020	2	05.07	17.18	12.14	+20
4021	2	05.07	16.19	12.14	+28	4022	2	05.07	21.22	12.14	+8
4023	2	05.07	20.23	12.14	+24	4024	2	05.07	04.06	13.15	+28
4025	2	05.07	00.02	13.15	−2	4026	2	05.07	01.03	13.15	−28
4027	2	05.07	12.14	13.15	−40	4028	2	05.07	08.10	13.15	−2
4029	2	05.07	09.11	13.15	−28	4030	2	05.07	17.18	13.15	−4
4031	2	05.07	16.19	13.15	+12	4032	2	05.07	21.22	13.15	−8
4034	2	05.07	04.06	08.10	−8	4035	2	05.07	00.02	08.10	−56
4036	2	05.07	01.03	08.10	−22	4037	2	05.07	12.14	08.10	−36
4038	2	05.07	13.15	08.10	+2	4040	2	05.07	17.18	08.10	+18
4041	2	05.07	16.19	08.10	+20	4042	2	05.07	21.22	08.10	+14
4043	2	05.07	20.23	08.10	+24	4044	2	05.07	04.06	09.11	−12
4045	2	05.07	00.02	09.11	−32	4046	2	05.07	01.03	09.11	−30
4047	2	05.07	12.14	09.11	+2	4048	2	05.07	13.15	09.11	−12
4050	2	05.07	17.18	09.11	−6	4051	2	05.07	16.19	09.11	+4
4052	2	05.07	21.22	09.11	−2	4054	2	05.07	04.06	17.18	+8
4055	2	05.07	00.02	17.18	+10	4056	2	05.07	01.03	17.18	−10
4057	2	05.07	12.14	17.18	+28	4058	2	05.07	13.15	17.18	+16
4059	2	05.07	08.10	17.18	+16	4060	2	05.07	09.11	17.18	+12
4061	2	05.07	16.19	17.18	−12	4062	2	05.07	21.22	17.18	−56
4063	2	05.07	20.23	17.18	−20	4064	2	05.07	04.06	16.19	+2
4065	2	05.07	00.02	16.19	−8	4066	2	05.07	01.03	16.19	−20
4067	2	05.07	12.14	16.19	+40	4068	2	05.07	13.15	16.19	+28
4069	2	05.07	08.10	16.19	+20	4070	2	05.07	09.11	16.19	+8
4071	2	05.07	17.18	16.19	−8	4072	2	05.07	21.22	16.19	−16
4073	2	05.07	20.23	16.19	−52	4074	2	05.07	04.06	21.22	+8

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
4075	2	05.07	00.02	21.22	+18	4076	2	05.07	01.03	21.22	+2
4077	2	05.07	12.14	21.22	+8	4078	2	05.07	13.15	21.22	−8
4079	2	05.07	08.10	21.22	+4	4080	2	05.07	09.11	21.22	−4
4081	2	05.07	17.18	21.22	−36	4082	2	05.07	16.19	21.22	−12
4083	2	05.07	20.23	21.22	+8	4084	2	05.07	04.06	20.23	+2
4085	2	05.07	00.02	20.23	+8	4086	2	05.07	01.03	20.23	−8
4087	2	05.07	12.14	20.23	+20	4088	2	05.07	13.15	20.23	+12
4089	2	05.07	08.10	20.23	+8	4091	2	05.07	17.18	20.23	−4
4092	2	05.07	16.19	20.23	−52	4105	2	04.06	01.03	00.02	+20
4106	2	04.06	12.14	00.02	−10	4107	2	04.06	13.15	00.02	−6
4108	2	04.06	08.10	00.02	−2	4109	2	04.06	09.11	00.02	+2
4110	2	04.06	17.18	00.02	+46	4111	2	04.06	16.19	00.02	+26
4112	2	04.06	21.22	00.02	+42	4113	2	04.06	20.23	00.02	+30
4115	2	04.06	00.02	01.03	+10	4116	2	04.06	12.14	01.03	+40
4117	2	04.06	13.15	01.03	+36	4118	2	04.06	08.10	01.03	+4
4119	2	04.06	09.11	01.03	+8	4120	2	04.06	17.18	01.03	+8
4121	2	04.06	16.19	01.03	+14	4122	2	04.06	21.22	01.03	+8
4123	2	04.06	20.23	01.03	+14	4125	2	04.06	00.02	12.14	−42
4126	2	04.06	01.03	12.14	+2	4127	2	04.06	13.15	12.14	−28
4128	2	04.06	08.10	12.14	+12	4129	2	04.06	09.11	12.14	−56
4130	2	04.06	17.18	12.14	+14	4132	2	04.06	21.22	12.14	+6
4133	2	04.06	20.23	12.14	−8	4135	2	04.06	00.02	13.15	−10
4136	2	04.06	01.03	13.15	+10	4137	2	04.06	12.14	13.15	−40
4138	2	04.06	08.10	13.15	−44	4140	2	04.06	17.18	13.15	−14
4141	2	04.06	16.19	13.15	−12	4142	2	04.06	21.22	13.15	−22
4143	2	04.06	20.23	13.15	−20	4145	2	04.06	00.02	08.10	+24
4146	2	04.06	01.03	08.10	−8	4147	2	04.06	12.14	08.10	+16
4148	2	04.06	13.15	08.10	−40	4150	2	04.06	17.18	08.10	−24
4151	2	04.06	16.19	08.10	−12	4152	2	04.06	21.22	08.10	−24
4153	2	04.06	20.23	08.10	−12	4157	2	04.06	12.14	09.11	−64
4158	2	04.06	13.15	09.11	+16	4160	2	04.06	17.18	09.11	+4
4162	2	04.06	21.22	09.11	+4	4165	2	04.06	00.02	17.18	−4
4166	2	04.06	01.03	17.18	−20	4167	2	04.06	12.14	17.18	+40
4168	2	04.06	13.15	17.18	+24	4170	2	04.06	09.11	17.18	+16
4171	2	04.06	16.19	17.18	−20	4173	2	04.06	20.23	17.18	−32
4175	2	04.06	00.02	16.19	−16	4176	2	04.06	01.03	16.19	−12
4177	2	04.06	12.14	16.19	+20	4178	2	04.06	13.15	16.19	+12
4179	2	04.06	08.10	16.19	+12	4180	2	04.06	09.11	16.19	+12
4181	2	04.06	17.18	16.19	−16	4182	2	04.06	21.22	16.19	−28
4183	2	04.06	20.23	16.19	+4	4185	2	04.06	00.02	21.22	+4
4186	2	04.06	01.03	21.22	−8	4187	2	04.06	12.14	21.22	+24
4188	2	04.06	13.15	21.22	+4	4189	2	04.06	08.10	21.22	−8
4190	2	04.06	09.11	21.22	+4	4191	2	04.06	17.18	21.22	+4
4192	2	04.06	16.19	21.22	−16	4193	2	04.06	20.23	21.22	−8
4197	2	04.06	12.14	20.23	−4	4198	2	04.06	13.15	20.23	−8
4199	2	04.06	08.10	20.23	−4	4201	2	04.06	17.18	20.23	−24
4202	2	04.06	16.19	20.23	+20	4226	2	00.02	12.14	01.03	+34
4227	2	00.02	13.15	01.03	+26	4228	2	00.02	08.10	01.03	+30
4229	2	00.02	09.11	01.03	+30	4230	2	00.02	17.18	01.03	−14
4231	2	00.02	16.19	01.03	−12	4232	2	00.02	21.22	01.03	−2
4233	2	00.02	20.23	01.03	−24	4237	2	00.02	13.15	12.14	+12
4238	2	00.02	08.10	12.14	−12	4239	2	00.02	09.11	12.14	−42
4240	2	00.02	17.18	12.14	+40	4241	2	00.02	16.19	12.14	+20
4242	2	00.02	21.22	12.14	+36	4243	2	00.02	20.23	12.14	+24
4248	2	00.02	08.10	13.15	−30	4250	2	00.02	17.18	13.15	−12
4251	2	00.02	16.19	13.15	−8	4252	2	00.02	21.22	13.15	−8
4253	2	00.02	20.23	13.15	−12	4257	2	00.02	12.14	08.10	−24
4258	2	00.02	13.15	08.10	−18	4260	2	00.02	17.18	08.10	+14
4261	2	00.02	16.19	08.10	+8	4262	2	00.02	21.22	08.10	+10
4263	2	00.02	20.23	08.10	+12	4267	2	00.02	12.14	09.11	−42
4270	2	00.02	17.18	09.11	+18	4271	2	00.02	16.19	09.11	+4
4272	2	00.02	21.22	09.11	+22	4277	2	00.02	12.14	17.18	+16
4278	2	00.02	13.15	17.18	−8	4279	2	00.02	08.10	17.18	−4
4280	2	00.02	09.11	17.18	+4	4281	2	00.02	16.19	17.18	−4
4282	2	00.02	21.22	17.18	−4	4283	2	00.02	20.23	17.18	+12
4287	2	00.02	12.14	16.19	+8	4288	2	00.02	13.15	16.19	−8
4289	2	00.02	08.10	16.19	+4	4290	2	00.02	09.11	16.19	−4
4291	2	00.02	17.18	16.19	+8	4292	2	00.02	21.22	16.19	+8

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
4293	2	00.02	20.23	16.19	+16	4297	2	00.02	12.14	21.22	+16
4298	2	00.02	13.15	21.22	−12	4299	2	00.02	08.10	21.22	−4
4302	2	00.02	16.19	21.22	+4	4307	2	00.02	12.14	20.23	+8
4308	2	00.02	13.15	20.23	−4	4309	2	00.02	08.10	20.23	+4
4311	2	00.02	17.18	20.23	+4	4347	2	01.03	13.15	12.14	+12
4348	2	01.03	08.10	12.14	+12	4350	2	01.03	17.18	12.14	−2
4352	2	01.03	21.22	12.14	−2	4358	2	01.03	08.10	13.15	−12
4360	2	01.03	17.18	13.15	−2	4362	2	01.03	21.22	13.15	−2
4387	2	01.03	12.14	17.18	+4	4389	2	01.03	08.10	17.18	−4
4391	2	01.03	16.19	17.18	+4	4392	2	01.03	21.22	17.18	+12
4397	2	01.03	12.14	16.19	+4	4399	2	01.03	08.10	16.19	+4
4402	2	01.03	21.22	16.19	+12	4407	2	01.03	12.14	21.22	+8
4470	2	12.14	17.18	13.15	+12	4471	2	12.14	16.19	13.15	+12
4472	2	12.14	21.22	13.15	+12	4473	2	12.14	20.23	13.15	+12
4480	2	12.14	17.18	08.10	+12	4481	2	12.14	16.19	08.10	−4
4482	2	12.14	21.22	08.10	+12	4483	2	12.14	20.23	08.10	−4
4491	2	12.14	16.19	09.11	+8	4492	2	12.14	21.22	09.11	+8
4498	2	12.14	13.15	17.18	−16	4499	2	12.14	08.10	17.18	−16
4500	2	12.14	09.11	17.18	−16	4501	2	12.14	16.19	17.18	+12
4502	2	12.14	21.22	17.18	+12	4503	2	12.14	20.23	17.18	+16
4509	2	12.14	08.10	16.19	−16	4510	2	12.14	09.11	16.19	−16
4511	2	12.14	17.18	16.19	+8	4512	2	12.14	21.22	16.19	+12
4513	2	12.14	20.23	16.19	+8	4518	2	12.14	13.15	21.22	−16
4521	2	12.14	17.18	21.22	+8	4522	2	12.14	16.19	21.22	+8
4523	2	12.14	20.23	21.22	+8	4531	2	12.14	17.18	20.23	+16
4532	2	12.14	16.19	20.23	+8	4590	2	13.15	17.18	08.10	−4
4591	2	13.15	16.19	08.10	−12	4592	2	13.15	21.22	08.10	−12
4593	2	13.15	20.23	08.10	−4	4600	2	13.15	17.18	09.11	−16
4602	2	13.15	21.22	09.11	−16	4609	2	13.15	08.10	17.18	−8
4610	2	13.15	09.11	17.18	−32	4611	2	13.15	16.19	17.18	+4
4612	2	13.15	21.22	17.18	+4	4613	2	13.15	20.23	17.18	+8
4619	2	13.15	08.10	16.19	−24	4622	2	13.15	21.22	16.19	+4
4632	2	13.15	16.19	21.22	+8	4633	2	13.15	20.23	21.22	−8
4721	2	08.10	16.19	17.18	+4	4722	2	08.10	21.22	17.18	+4
4723	2	08.10	20.23	17.18	+8	4731	2	08.10	17.18	16.19	+8
4732	2	08.10	21.22	16.19	−4	4831	2	09.11	16.19	17.18	+12
4832	2	09.11	21.22	17.18	+4	4842	2	09.11	21.22	16.19	+4
5304	2	05.14	03.09	00.10	−24	5305	2	05.14	06.13	00.10	−12
5306	2	05.14	00.10	03.09	−12	5307	2	05.14	06.13	03.09	+20
5308	2	05.14	00.10	06.13	−10	5313	2	00.10	06.13	03.09	−26
5315	2	00.10	03.09	06.13	−2	5328	2	05.15	03.08	00.11	−24
5329	2	05.15	06.12	00.11	−12	5330	2	05.15	00.11	03.08	−12
5331	2	05.15	06.12	03.08	+20	5332	2	05.15	00.11	06.12	−10
5337	2	00.11	06.12	03.08	−26	5339	2	00.11	03.08	06.12	−2
5352	2	05.08	03.15	00.12	−12	5353	2	05.08	06.11	00.12	−8
5354	2	05.08	00.12	03.15	−8	5355	2	05.08	06.11	03.15	+8
5356	2	05.08	00.12	06.11	+2	5357	2	05.08	03.15	06.11	+4
5361	2	00.12	06.11	03.15	−34	5363	2	00.12	03.15	06.11	−10
5376	2	05.09	03.14	00.13	−12	5377	2	05.09	06.10	00.13	−8
5378	2	05.09	00.13	03.14	−8	5379	2	05.09	06.10	03.14	+8
5380	2	05.09	00.13	06.10	+2	5381	2	05.09	03.14	06.10	+4
5385	2	00.13	06.10	03.14	−34	5387	2	00.13	03.14	06.10	−10
5400	2	05.10	03.13	00.14	−28	5401	2	05.10	06.09	00.14	−20
5402	2	05.10	00.14	03.13	−20	5403	2	05.10	06.09	03.13	+20
5404	2	05.10	00.14	06.09	−4	5405	2	05.10	03.13	06.09	+4
5409	2	00.14	06.09	03.13	−40	5411	2	00.14	03.13	06.09	−8
5424	2	05.11	03.12	00.15	−28	5425	2	05.11	06.08	00.15	−20
5426	2	05.11	00.15	03.12	−20	5427	2	05.11	06.08	03.12	+20
5428	2	05.11	00.15	06.08	−4	5429	2	05.11	03.12	06.08	+4
5433	2	00.15	06.08	03.12	−40	5435	2	00.15	03.12	06.08	−8
5496	2	05.18	03.21	00.23	−28	5497	2	05.18	06.16	00.23	+16
5498	2	05.18	00.23	03.21	−28	5499	2	05.18	06.16	03.21	+4
5500	2	05.18	00.23	06.16	−4	5501	2	05.18	03.21	06.16	+12
5505	2	00.23	06.16	03.21	−8	5507	2	00.23	03.21	06.16	−4
5520	2	05.19	03.20	00.22	−36	5521	2	05.19	06.17	00.22	+8
5522	2	05.19	00.22	03.20	−16	5523	2	05.19	06.17	03.20	+8
5524	2	05.19	00.22	06.17	+4	5525	2	05.19	03.20	06.17	+20
5529	2	00.22	06.17	03.20	+4	5531	2	00.22	03.20	06.17	+4

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
5544	2	05.22	03.17	00.19	−24	5545	2	05.22	06.20	00.19	+20
5546	2	05.22	00.19	03.17	−24	5548	2	05.22	00.19	06.20	−4
5549	2	05.22	03.17	06.20	+12	5553	2	00.19	06.20	03.17	−8
5555	2	00.19	03.17	06.20	−8	5568	2	05.23	03.16	00.18	−32
5569	2	05.23	06.21	00.18	+12	5570	2	05.23	00.18	03.16	−12
5571	2	05.23	06.21	03.16	+12	5572	2	05.23	00.18	06.21	+4
5573	2	05.23	03.16	06.21	+20	5577	2	00.18	06.21	03.16	+4
5579	2	00.18	03.16	06.21	+8	5592	2	05.21	03.18	00.16	−28
5593	2	05.21	06.23	00.16	+28	5594	2	05.21	00.16	03.18	−24
5596	2	05.21	00.16	06.23	+8	5597	2	05.21	03.18	06.23	+24
5601	2	00.16	06.23	03.18	+12	5603	2	00.16	03.18	06.23	−4
5616	2	05.20	03.19	00.17	−4	5617	2	05.20	06.22	00.17	+4
5618	2	05.20	00.17	03.19	−4	5619	2	05.20	06.22	03.19	+12
5627	2	00.17	03.19	06.22	+4	5640	2	05.17	03.22	00.20	−24
5641	2	05.17	06.19	00.20	+24	5642	2	05.17	00.20	03.22	−20
5643	2	05.17	06.19	03.22	−4	5644	2	05.17	00.20	06.19	+8
5645	2	05.17	03.22	06.19	+24	5649	2	00.20	06.19	03.22	+12
5688	2	04.04	02.02	01.01	+8	5689	2	04.04	07.07	01.01	−10
5690	2	04.04	01.01	02.02	+2	5691	2	04.04	07.07	02.02	−2
5692	2	04.04	01.01	07.07	−18	5693	2	04.04	02.02	07.07	+28
5712	2	04.14	02.09	01.10	−24	5713	2	04.14	07.13	01.10	+20
5714	2	04.14	01.10	02.09	−18	5715	2	04.14	07.13	02.09	+2
5716	2	04.14	01.10	07.13	+12	5717	2	04.14	02.09	07.13	−10
5736	2	04.15	02.08	01.11	−24	5737	2	04.15	07.12	01.11	+20
5738	2	04.15	01.11	02.08	−18	5739	2	04.15	07.12	02.08	+2
5740	2	04.15	01.11	07.12	+12	5741	2	04.15	02.08	07.12	−10
5760	2	04.08	02.15	01.12	+4	5761	2	04.08	07.11	01.12	+8
5762	2	04.08	01.12	02.15	−6	5763	2	04.08	07.11	02.15	−2
5764	2	04.08	01.12	07.11	+16	5765	2	04.08	02.15	07.11	−30
5784	2	04.09	02.14	01.13	+4	5785	2	04.09	07.10	01.13	+8
5786	2	04.09	01.13	02.14	−6	5787	2	04.09	07.10	02.14	−2
5788	2	04.09	01.13	07.10	+16	5789	2	04.09	02.14	07.10	−30
5808	2	04.10	02.13	01.14	−12	5809	2	04.10	07.09	01.14	+28
5810	2	04.10	01.14	02.13	−12	5811	2	04.10	07.09	02.13	−4
5812	2	04.10	01.14	07.09	+20	5813	2	04.10	02.13	07.09	−20
5832	2	04.11	02.12	01.15	−12	5833	2	04.11	07.08	01.15	+28
5834	2	04.11	01.15	02.12	−12	5835	2	04.11	07.08	02.12	−4
5836	2	04.11	01.15	07.08	+20	5837	2	04.11	02.12	07.08	−20
5904	2	04.18	02.21	01.23	−8	5905	2	04.18	07.16	01.23	−4
5907	2	04.18	07.16	02.21	+8	5908	2	04.18	01.23	07.16	+4
5909	2	04.18	02.21	07.16	−12	5928	2	04.19	02.20	01.22	−4
5929	2	04.19	07.17	01.22	+8	5930	2	04.19	01.22	02.20	−12
5931	2	04.19	07.17	02.20	+4	5932	2	04.19	01.22	07.17	+8
5933	2	04.19	02.20	07.17	−16	5952	2	04.22	02.17	01.19	−8
5953	2	04.22	07.20	01.19	+4	5955	2	04.22	07.20	02.17	+8
5956	2	04.22	01.19	07.20	+4	5957	2	04.22	02.17	07.20	−12
5976	2	04.23	02.16	01.18	+4	5977	2	04.23	07.21	01.18	+8
5978	2	04.23	01.18	02.16	−4	5979	2	04.23	07.21	02.16	+4
5980	2	04.23	01.18	07.21	+8	5981	2	04.23	02.16	07.21	−16
6000	2	04.21	02.18	01.16	+4	6001	2	04.21	07.23	01.16	−4
6002	2	04.21	01.16	02.18	+4	6003	2	04.21	07.23	02.18	+4
6004	2	04.21	01.16	07.23	+4	6005	2	04.21	02.18	07.23	−20
6027	2	04.20	07.22	02.19	+8	6048	2	04.17	02.22	01.20	+12
6049	2	04.17	07.19	01.20	−12	6050	2	04.17	01.20	02.22	+12
6051	2	04.17	07.19	02.22	−4	6052	2	04.17	01.20	07.19	+4
6053	2	04.17	02.22	07.19	−20	6120	2	14.18	10.23	09.21	+4
6121	2	14.18	13.16	09.21	−4	6122	2	14.18	09.21	10.23	+8
6124	2	14.18	09.21	13.16	−4	6125	2	14.18	10.23	13.16	+4
6129	2	09.21	13.16	10.23	−4	6144	2	14.19	10.22	09.20	+12
6145	2	14.19	13.17	09.20	−4	6146	2	14.19	09.20	10.22	+8
6148	2	14.19	09.20	13.17	−4	6149	2	14.19	10.22	13.17	+4
6153	2	09.20	13.17	10.22	+4	6170	2	14.22	09.17	10.19	+4
6171	2	14.22	13.20	10.19	−4	6172	2	14.22	09.17	13.20	−4
6173	2	14.22	10.19	13.20	+4	6177	2	09.17	13.20	10.19	−4
6179	2	09.17	10.19	13.20	+4	6192	2	14.23	10.18	09.16	+8
6193	2	14.23	13.21	09.16	−8	6194	2	14.23	09.16	10.18	+4
6195	2	14.23	13.21	10.18	−4	6196	2	14.23	09.16	13.21	−4
6197	2	14.23	10.18	13.21	+4	6201	2	09.16	13.21	10.18	+4

j	d	Z	K	O	c_j	j	d	Z	K	O	c_j
6203	2	09.16	10.18	13.21	-4	6216	2	14.21	10.16	09.18	+4
6217	2	14.21	13.23	09.18	-4	6218	2	14.21	09.18	10.16	+4
6219	2	14.21	13.23	10.16	-4	6227	2	09.18	10.16	13.23	+4
6240	2	14.20	10.17	09.19	+4	6241	2	14.20	13.22	09.19	-12
6242	2	14.20	09.19	10.17	+4	6243	2	14.20	13.22	10.17	-4
6251	2	09.19	10.17	13.22	-4	6336	2	15.18	11.23	08.21	+8
6337	2	15.18	12.16	08.21	-8	6338	2	15.18	08.21	11.23	+8
6340	2	15.18	08.21	12.16	-8	6341	2	15.18	11.23	12.16	+8
6360	2	15.19	11.22	08.20	+8	6362	2	15.19	08.20	11.22	+8
6384	2	15.22	11.19	08.17	+8	6385	2	15.22	12.20	08.17	-8
6386	2	15.22	08.17	11.19	+8	6387	2	15.22	12.20	11.19	-8
6388	2	15.22	08.17	12.20	-8	6389	2	15.22	11.19	12.20	+8
6432	2	15.21	11.16	08.18	+8	6433	2	15.21	12.23	08.18	-8
6434	2	15.21	08.18	11.16	+8	6435	2	15.21	12.23	11.16	-8
6457	2	15.20	12.22	08.19	-8	6576	3	05.14.15	03.03.04	00.18.22	+8
6578	3	05.08.09	03.03.04	00.19.23	+8	6579	3	05.08.09	00.19.23	03.03.04	+16
6580	3	05.10.11	04.06.06	00.19.23	+6	6581	3	05.10.11	00.19.23	04.06.06	+10
6582	3	05.10.11	00.16.20	01.03.06	-2	6583	3	05.10.11	01.03.06	00.16.20	+2
6584	3	05.12.13	04.06.06	00.18.22	-10	6585	3	05.12.13	00.18.22	04.06.06	-6
6586	3	05.12.13	00.17.21	01.03.06	-2	6587	3	05.12.13	01.03.06	00.17.21	-14
6612	3	05.18.22	00.14.15	01.03.03	+12	6613	3	05.18.22	00.12.13	01.06.06	+16
6614	3	05.18.22	01.03.03	00.14.15	+6	6616	3	05.19.23	00.08.09	01.03.03	-4
6617	3	05.19.23	00.10.11	01.06.06	+16	6618	3	05.19.23	01.03.03	00.08.09	+6
6619	3	05.19.23	01.06.06	00.10.11	+16	6620	3	05.17.21	03.04.06	00.14.15	+2
6621	3	05.17.21	00.14.15	03.04.06	+12	6622	3	05.16.20	03.04.06	00.08.09	-14
6623	3	05.16.20	00.08.09	03.04.06	-4	6648	3	02.05.05	03.14.15	06.16.20	-8
6649	3	02.05.05	03.08.09	06.17.21	+8	6650	3	02.05.05	06.17.21	03.08.09	-8
6651	3	02.05.05	06.16.20	03.14.15	+8	6652	3	00.05.07	03.10.11	06.18.22	+12
6653	3	00.05.07	03.12.13	06.19.23	-4	6654	3	00.05.07	06.18.22	03.10.11	-6
6655	3	00.05.07	06.19.23	03.12.13	-6	6656	3	00.00.02	03.10.11	06.17.21	+4
6657	3	00.00.02	03.12.13	06.16.20	+4	6658	3	00.00.02	06.17.21	03.10.11	-10
6659	3	00.00.02	06.16.20	03.12.13	+6	6684	3	05.05.07	03.17.21	06.08.09	+10
6685	3	05.05.07	03.16.20	06.14.15	-6	6686	3	05.05.07	06.14.15	03.16.20	-4
6687	3	05.05.07	06.08.09	03.17.21	-4	6688	3	00.02.05	03.18.22	06.08.09	+14
6689	3	00.02.05	03.19.23	06.14.15	+14	6690	3	00.02.05	06.14.15	03.19.23	-4
6691	3	00.02.05	06.08.09	03.18.22	+12	6692	3	00.00.07	03.17.21	06.10.11	+16
6694	3	00.00.07	06.10.11	03.17.21	+16	6721	3	02.02.05	04.08.09	01.18.22	-16
6722	3	02.02.05	01.18.22	04.08.09	-8	6723	3	02.02.05	01.19.23	04.14.15	-8
6724	3	05.07.07	04.10.11	01.18.22	-16	6867	3	04.06.13	10.16.20	00.07.09	+8
6869	3	00.02.13	10.18.22	01.06.09	-8	6891	3	04.06.12	11.16.20	00.07.08	+8
6893	3	00.02.12	11.18.22	01.06.08	-8	6915	3	04.06.11	12.17.21	00.07.15	+8
6917	3	00.02.11	12.19.23	01.06.15	-8	6939	3	04.06.10	13.17.21	00.07.14	+8
6941	3	00.02.10	13.19.23	01.06.14	-8						

A.5 All 1,414 rational coefficients of the cubic logarithm polynomial

Each row (a, b, c, p) contributes exactly $p\,\ell_a(q_1)\ell_b(q_1)\ell_c(q_1)$ to $\mathcal{P}(q_1)$ (no factorial and no implicit symmetrization).

a	b	c	p_{abc}	a	b	c	p_{abc}
0	0	0	-88/3	0	0	1	11
0	0	2	6	0	0	3	-43
0	0	4	32	0	0	5	11
0	0	6	152	0	0	7	-46
0	0	9	-2	0	0	10	2
0	0	11	-20	0	0	12	-12
0	0	14	8	0	0	17	16
0	0	19	8	0	1	1	-20
0	1	2	8	0	1	3	-24
0	1	4	11	0	1	5	35
0	1	6	-14	0	1	7	2
0	1	8	-18	0	1	9	10
0	1	10	-4	0	1	11	-8
0	1	12	-8	0	1	15	-32
0	1	16	4	0	1	17	16

a	b	c	p_{abc}	a	b	c	p_{abc}
0	1	18	10	0	2	2	16
0	2	3	-22	0	2	4	81/2
0	2	5	-9/2	0	2	6	27
0	2	7	-66	0	2	8	8
0	2	9	-8	0	2	11	-34
0	2	12	-48	0	2	13	27
0	2	14	5	0	2	17	88
0	2	19	-24	0	3	3	70
0	3	4	-165	0	3	5	-47
0	3	6	-16	0	3	7	310
0	3	8	-24	0	3	9	-8
0	3	10	-16	0	3	11	98
0	3	12	118	0	3	13	-52
0	3	14	-16	0	3	15	-34
0	3	16	-18	0	3	17	-208
0	3	18	32	0	3	19	44
0	4	4	275/4	0	4	5	57/2
0	4	6	127	0	4	7	-245
0	4	8	-2	0	4	9	-18
0	4	10	-24	0	4	11	-215/2
0	4	12	-199/2	0	4	13	55/2
0	4	14	71/2	0	4	15	36
0	4	16	32	0	4	17	198
0	4	18	-12	0	4	19	16
0	5	5	7/4	0	5	6	-53
0	5	7	-39	0	5	8	46
0	5	9	-2	0	5	10	48
0	5	11	-21/2	0	5	12	-5/2
0	5	13	9/2	0	5	14	25/2
0	5	15	4	0	5	17	10
0	5	19	8	0	6	6	-113
0	6	7	-338	0	6	8	20
0	6	9	60	0	6	10	8
0	6	11	-141	0	6	12	-209
0	6	13	77	0	6	14	-23
0	6	15	76	0	6	16	36
0	6	17	340	0	6	18	-72
0	6	19	-112	0	7	7	214
0	7	8	12	0	7	10	28
0	7	11	194	0	7	12	194
0	7	13	-30	0	7	14	-30
0	7	15	-76	0	7	16	-60
0	7	17	-352	0	7	18	24
0	7	19	-16	0	8	8	-30
0	8	9	20	0	8	10	-4
0	8	11	-8	0	8	12	-8
0	8	15	-42	0	8	16	-18
0	8	17	16	0	8	18	40
0	9	9	-50	0	9	10	4
0	9	11	8	0	9	12	8
0	9	15	2	0	9	16	10
0	9	17	-16	0	9	18	28
0	10	10	-8	0	10	15	-44
0	10	16	-28	0	10	18	12
0	11	11	28	0	11	12	176
0	11	13	-29	0	11	14	-7
0	11	17	-230	0	11	19	40
0	12	12	32	0	12	13	1
0	12	14	-5	0	12	17	-214
0	12	19	-20	0	13	13	-47
0	13	14	50	0	13	17	18
0	13	19	72	0	14	14	-19
0	14	17	34	0	14	19	-20
0	15	15	4	0	15	16	64
0	15	18	-4	0	16	16	-4

a	b	c	p_{abc}	a	b	c	p_{abc}
0	16	18	-20	0	17	17	192
0	17	19	32	0	18	18	-24
0	19	19	-56	1	1	1	4/3
1	1	2	4	1	1	3	57
1	1	4	-123/2	1	1	5	-49/2
1	1	6	-61	1	1	7	96
1	1	8	3	1	1	9	63
1	1	10	35	1	1	11	-8
1	1	12	-16	1	1	13	8
1	1	15	-14	1	1	16	-107
1	1	17	16	1	1	18	44
1	1	19	-8	1	2	2	-27/4
1	2	3	44	1	2	4	-49/2
1	2	5	7/2	1	2	6	-112
1	2	7	64	1	2	8	4
1	2	9	-12	1	2	10	16
1	2	11	75/2	1	2	12	35/2
1	2	13	69/2	1	2	14	13/2
1	2	15	-8	1	2	16	-8
1	2	17	-39	1	2	18	12
1	2	19	-52	1	3	3	5
1	3	4	21	1	3	5	-59
1	3	6	84	1	3	7	-74
1	3	8	110	1	3	9	20
1	3	10	34	1	3	11	-88
1	3	12	-52	1	3	13	-56
1	3	14	-4	1	3	15	26
1	3	16	-82	1	3	17	88
1	3	18	-46	1	3	19	88
1	4	4	-13/2	1	4	5	31
1	4	6	-68	1	4	7	42
1	4	8	-73	1	4	9	101
1	4	10	-129	1	4	11	121/2
1	4	12	145/2	1	4	13	7/2
1	4	14	15/2	1	4	15	22
1	4	16	101	1	4	17	-113
1	4	18	-108	1	4	19	8
1	5	5	47/2	1	5	6	12
1	5	7	-10	1	5	8	-51
1	5	9	-23	1	5	10	7
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1	5	13	-1/2	1	5	14	-1/2
1	5	15	-30	1	5	16	23
1	5	17	9	1	5	18	42
1	6	6	-77	1	6	7	90
1	6	8	-132	1	6	9	-62
1	6	10	-88	1	6	11	200
1	6	12	128	1	6	13	112
1	6	14	8	1	6	15	18
1	6	16	240	1	6	17	-234
1	6	18	-38	1	6	19	-176
1	7	7	-36	1	7	8	126
1	7	9	-172	1	7	10	276
1	7	11	-136	1	7	12	-160
1	7	13	-8	1	7	14	-16
1	7	15	-100	1	7	16	-214
1	7	17	266	1	7	18	230
1	7	19	-16	1	8	8	23
1	8	9	100	1	8	10	-20
1	8	11	8	1	8	12	-8
1	8	13	24	1	8	14	8
1	8	15	-14	1	8	16	-144
1	8	17	16	1	8	18	32
1	8	19	-16	1	9	9	-29
1	9	10	-56	1	9	11	28

a	b	c	p_{abc}	a	b	c	p_{abc}
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1	9	16	76	1	9	17	-32
1	9	18	-152	1	10	10	121
1	10	11	-28	1	10	12	-44
1	10	13	20	1	10	14	4
1	10	15	-138	1	10	16	-260
1	10	17	64	1	10	18	242
1	10	19	-16	1	11	11	-187/4
1	11	12	-35/2	1	11	13	-117/2
1	11	14	-61/2	1	11	15	20
1	11	16	28	1	11	17	71
1	11	18	-44	1	11	19	100
1	12	12	-123/4	1	12	13	-45/2
1	12	14	-5/2	1	12	15	20
1	12	16	44	1	12	17	79
1	12	18	-44	1	12	19	4
1	13	13	-171/4	1	13	14	29/2
1	13	15	-4	1	13	16	-4
1	13	17	17	1	13	18	-4
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1	14	17	9	1	14	18	-4
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1	15	16	28	1	15	17	-32
1	15	18	-146	1	16	16	123
1	16	17	-48	1	16	18	-120
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1	19	19	-96	2	2	2	17/12
2	2	3	-5/2	2	2	4	63/4
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2	2	9	47/4	2	2	10	-3/2
2	2	11	23/4	2	2	12	-15/4
2	2	13	33/2	2	2	14	24
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2	2	19	-1	2	3	3	-1
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2	4	10	-17	2	4	11	-63/2
2	4	12	-77/2	2	4	13	17/2
2	4	14	61	2	4	15	-10
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2	4	18	-32	2	4	19	111/2
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2	5	7	12	2	5	8	13/2
2	5	9	-5/2	2	5	11	4
2	5	12	1	2	5	13	2
2	5	14	-7/2	2	5	17	-10
2	5	19	-1/2	2	6	6	-296
2	6	7	260	2	6	8	-68
2	6	9	120	2	6	10	-86
2	6	11	252	2	6	12	216
2	6	13	25	2	6	14	130
2	6	15	-8	2	6	16	104

a	b	c	p_{abc}	a	b	c	p_{abc}
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2	7	12	-18	2	7	13	-5
2	7	14	-111	2	7	15	20
2	7	16	-44	2	7	17	8
2	7	18	64	2	7	19	-90
2	8	8	-3	2	8	9	-8
2	8	10	12	2	8	11	47/2
2	8	12	-9/2	2	8	13	61/2
2	8	14	13/2	2	8	15	-4
2	8	16	-4	2	8	17	-11
2	8	18	8	2	8	19	-40
2	9	9	7	2	9	10	-8
2	9	11	-83/2	2	9	12	-67/2
2	9	13	-61/2	2	9	14	-13/2
2	9	17	55	2	9	18	-4
2	9	19	52	2	10	10	6
2	10	11	27	2	10	12	9
2	10	13	34	2	10	15	4
2	10	16	-16	2	10	17	-22
2	10	18	4	2	10	19	-56
2	11	11	-167/4	2	11	12	-49/2
2	11	13	-57	2	11	14	-74
2	11	15	8	2	11	16	-32
2	11	17	34	2	11	18	14
2	11	19	42	2	12	12	-3/4
2	12	13	-42	2	12	14	-63
2	12	15	2	2	12	16	-10
2	12	17	-8	2	12	18	48
2	12	19	4	2	13	13	21/4
2	13	14	-13/2	2	13	15	-2
2	13	16	-42	2	13	17	81
2	13	18	-2	2	14	14	17/4
2	14	17	123	2	14	19	30
2	15	15	-10	2	15	18	12
2	16	16	10	2	16	17	24
2	16	18	-8	2	16	19	68
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3	3	10	18	3	3	11	57
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3	4	10	37	3	4	11	-111/2
3	4	12	-111/2	3	4	13	19/2
3	4	14	-219/2	3	4	15	-31
3	4	16	-129	3	4	17	-25
3	4	18	152	3	4	19	-105
3	5	5	-11/2	3	5	6	219
3	5	7	16	3	5	8	-99
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3	5	11	9/2	3	5	12	9/2
3	5	13	-29/2	3	5	14	-19/2
3	5	15	17	3	5	16	35
3	5	17	19	3	5	18	18

a	b	c	p_{abc}	a	b	c	p_{abc}
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3	6	9	-210	3	6	10	26
3	6	11	-676	3	6	12	-736
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3	6	15	216	3	6	16	-84
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3	6	19	-242	3	7	7	135
3	7	8	42	3	7	9	90
3	7	10	-42	3	7	11	333
3	7	12	321	3	7	13	-37
3	7	14	195	3	7	15	-24
3	7	16	208	3	7	17	-308
3	7	18	-252	3	7	19	168
3	8	8	107	3	8	9	-8
3	8	10	104	3	8	11	-50
3	8	12	-2	3	8	13	-46
3	8	14	-6	3	8	15	-56
3	8	16	-78	3	8	17	40
3	8	18	-72	3	8	19	64
3	9	9	-1	3	9	10	24
3	9	11	92	3	9	12	68
3	9	13	52	3	9	14	-4
3	9	15	10	3	9	16	-20
3	9	17	-96	3	9	18	-12
3	9	19	-96	3	10	10	24
3	10	11	-81	3	10	12	-45
3	10	13	-55	3	10	14	13
3	10	15	-84	3	10	16	-32
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3	10	19	88	3	11	11	263/2
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3	11	14	129	3	11	15	-28
3	11	16	84	3	11	17	-296
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3	13	13	-89/2	3	13	14	53
3	13	15	12	3	13	16	92
3	13	17	-104	3	13	18	-2
3	13	19	128	3	14	14	-5/2
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3	14	18	18	3	14	19	-68
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3	16	17	-64	3	16	18	98
3	16	19	-136	3	17	17	74
3	17	18	136	3	17	19	16
3	18	18	-30	3	18	19	-8
3	19	19	-104	4	4	4	761/12
4	4	5	37/4	4	4	6	-181/4
4	4	7	-947/4	4	4	8	-49/4
4	4	9	-265/4	4	4	10	-11
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4	4	13	155/4	4	4	14	203/4
4	4	15	42	4	4	16	84
4	4	17	117	4	4	18	-12
4	4	19	-14	4	5	5	69/4
4	5	6	-217/2	4	5	7	-83/2
4	5	8	153/2	4	5	9	1/2
4	5	10	11	4	5	11	-1/2
4	5	12	-9/2	4	5	13	7

a	b	c	p_{abc}	a	b	c	p_{abc}
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4	6	8	-10	4	6	9	60
4	6	10	-22	4	6	11	911/2
4	6	12	887/2	4	6	13	-239/2
4	6	14	237/2	4	6	15	-28
4	6	16	100	4	6	17	-632
4	6	18	-136	4	6	19	198
4	7	7	215	4	7	8	60
4	7	9	192	4	7	10	4
4	7	11	-203	4	7	12	-85
4	7	13	-97	4	7	14	-143
4	7	15	-108	4	7	16	-260
4	7	17	-120	4	7	18	60
4	7	19	64	4	8	8	-56
4	8	9	125	4	8	10	-87
4	8	11	31/2	4	8	12	31/2
4	8	13	33/2	4	8	14	33/2
4	8	15	59	4	8	16	-31
4	8	17	-23	4	8	18	-50
4	8	19	-8	4	9	9	-126
4	9	10	213	4	9	11	-107/2
4	9	12	-139/2	4	9	13	-1/2
4	9	14	15/2	4	9	15	-216
4	9	16	-82	4	9	17	103
4	9	18	268	4	9	19	-16
4	10	10	-87	4	10	11	47
4	10	12	67	4	10	13	-8
4	10	14	-12	4	10	15	79
4	10	16	99	4	10	17	-104
4	10	18	-186	4	10	19	16
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4	11	13	47/2	4	11	14	-79
4	11	15	38	4	11	16	-30
4	11	17	88	4	11	18	68
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4	12	17	22	4	12	18	88
4	12	19	-7/2	4	13	13	289/4
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4	13	16	-2	4	13	17	126
4	13	18	-12	4	13	19	-337/2
4	14	14	63/4	4	14	15	14
4	14	16	-6	4	14	17	160
4	14	18	-32	4	14	19	99/2
4	15	15	-40	4	15	16	136
4	15	17	-32	4	15	18	30
4	15	19	-16	4	16	16	-24
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4	16	19	-8	4	17	17	144
4	17	18	-136	4	17	19	-32
4	18	18	-66	4	18	19	16
4	19	19	127	5	5	5	-9/4
5	5	6	-133/4	5	5	7	-139/4
5	5	8	139/4	5	5	9	-53/4
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5	5	12	-1	5	5	13	11/4
5	5	14	19/4	5	5	15	2
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5	6	9	20	5	6	10	-4
5	6	11	11/2	5	6	12	-13/2

a	b	c	p_{abc}	a	b	c	p_{abc}
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5	6	19	-14	5	7	7	-1
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5	7	12	10	5	7	13	-18
5	7	14	-16	5	7	15	-14
5	7	16	-14	5	7	17	32
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5	8	10	-61	5	8	11	-5/2
5	8	12	-5/2	5	8	13	5/2
5	8	14	5/2	5	8	15	13
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5	8	18	26	5	9	9	17
5	9	10	-65	5	9	11	-3/2
5	9	12	-3/2	5	9	13	-1/2
5	9	14	-1/2	5	9	15	28
5	9	16	42	5	9	17	-7
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5	10	13	-1	5	10	14	-1
5	10	15	-11	5	10	16	-7
5	10	17	-16	5	10	18	66
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5	11	17	3	5	11	19	-7/2
5	12	12	-1/2	5	12	13	-11
5	12	14	13/2	5	12	17	-3
5	12	19	1/2	5	13	13	-21/2
5	13	14	-13/2	5	13	17	11
5	13	19	15/2	5	14	14	-7
5	14	17	17	5	14	19	-21/2
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6	6	18	-96	6	6	19	116
6	7	7	-978	6	7	8	-24
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6	7	13	206	6	7	14	-274
6	7	15	248	6	7	16	-104
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6	7	19	-328	6	8	8	-104
6	8	9	-98	6	8	10	-166
6	8	11	124	6	8	12	28
6	8	13	92	6	8	14	12
6	8	15	180	6	8	16	180
6	8	17	-126	6	8	18	52
6	8	19	-128	6	9	9	110
6	9	10	-2	6	9	11	-196
6	9	12	-148	6	9	13	-104
6	9	14	8	6	9	15	-58
6	9	16	-90	6	9	17	238
6	9	18	108	6	9	19	192
6	10	10	-36	6	10	11	136
6	10	12	64	6	10	13	110

a	b	c	p_{abc}	a	b	c	p_{abc}
6	10	14	-26	6	10	15	274
6	10	16	210	6	10	17	-160
6	10	18	-160	6	10	19	-176
6	11	11	-396	6	11	12	-1062
6	11	13	25	6	11	14	-244
6	11	15	56	6	11	16	-168
6	11	17	1336	6	11	18	92
6	11	19	31	6	12	12	-274
6	12	13	-127	6	12	14	-108
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6	13	16	-184	6	13	17	104
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6	14	17	268	6	14	18	-36
6	14	19	177	6	15	15	-178
6	15	16	8	6	15	17	-32
6	15	18	16	6	15	19	-16
6	16	16	-106	6	16	17	128
6	16	18	64	6	16	19	272
6	17	17	-828	6	17	18	-272
6	17	19	-64	6	18	18	-36
6	18	19	16	6	19	19	74
7	7	7	-85/3	7	7	8	-43
7	7	9	-171	7	7	10	28
7	7	11	362	7	7	12	250
7	7	13	63	7	7	14	111
7	7	15	44	7	7	16	188
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7	7	19	-64	7	8	8	79
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7	8	11	-52	7	8	12	-52
7	8	13	-28	7	8	14	-28
7	8	15	-216	7	8	16	16
7	8	17	110	7	8	18	160
7	8	19	16	7	9	9	169
7	9	10	-410	7	9	11	132
7	9	12	164	7	9	14	-16
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7	9	17	-254	7	9	18	-452
7	9	19	32	7	10	10	184
7	10	11	-68	7	10	12	-108
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7	10	15	-226	7	10	16	-266
7	10	17	192	7	10	18	356
7	10	19	-32	7	11	11	185
7	11	12	518	7	11	13	-25
7	11	14	153	7	11	15	-76
7	11	16	60	7	11	17	-550
7	11	18	-136	7	11	19	126
7	12	12	139	7	12	13	47
7	12	14	125	7	12	15	-52
7	12	16	100	7	12	17	-414
7	12	18	-176	7	12	19	-14
7	13	13	-166	7	13	14	114
7	13	15	-4	7	13	16	4
7	13	17	-194	7	13	18	24
7	13	19	314	7	14	14	-50
7	14	15	-28	7	14	16	12
7	14	17	-250	7	14	18	64
7	14	19	-122	7	15	15	84
7	15	16	-160	7	15	17	64
7	15	19	32	7	16	16	60

a	b	c	p_{abc}	a	b	c	p_{abc}
7	16	17	-160	7	16	18	-144
7	16	19	16	7	17	17	72
7	17	18	272	7	17	19	80
7	18	18	60	7	18	19	-32
7	19	19	-212	8	8	8	12
8	8	9	51	8	8	10	-5
8	8	11	19	8	8	12	11
8	8	13	13	8	8	14	5
8	8	15	-37	8	8	16	-67
8	8	17	-12	8	8	19	-8
8	9	9	-9	8	9	10	6
8	9	11	4	8	9	12	12
8	9	13	-12	8	9	14	-4
8	9	15	130	8	9	16	34
8	9	17	-16	8	9	18	-194
8	10	10	85	8	10	11	-4
8	10	12	-20	8	10	13	28
8	10	14	12	8	10	15	-214
8	10	16	-102	8	10	17	48
8	10	18	186	8	10	19	-16
8	11	11	-163/4	8	11	12	53/2
8	11	13	-121/2	8	11	14	-65/2
8	11	15	8	8	11	16	4
8	11	17	23	8	11	18	-32
8	11	19	96	8	12	12	-67/4
8	12	13	-25/2	8	12	14	-17/2
8	12	15	8	8	12	16	20
8	12	17	7	8	12	18	-32
8	13	13	-187/4	8	13	14	53/2
8	13	16	-12	8	13	17	41
8	13	18	-8	8	13	19	72
8	14	14	-43/4	8	14	16	4
8	14	17	25	8	14	18	-8
8	14	19	-8	8	15	15	163
8	15	16	-30	8	15	17	-16
8	15	18	-190	8	16	16	63
8	16	17	-32	8	16	18	30
8	16	19	16	8	17	17	-28
8	17	18	32	8	17	19	-32
8	18	18	108	8	19	19	-56
9	9	9	-100/3	9	9	10	95
9	9	11	-17	9	9	12	-17
9	9	13	-3	9	9	14	-3
9	9	15	-172	9	9	16	-54
9	9	17	12	9	9	18	168
9	9	19	8	9	10	10	-149
9	10	11	40	9	10	12	40
9	10	15	264	9	10	16	200
9	10	17	-32	9	10	18	-330
9	11	11	111/4	9	11	12	87/2
9	11	13	89/2	9	11	14	41/2
9	11	15	-12	9	11	16	-20
9	11	17	-79	9	11	18	36
9	11	19	-92	9	12	12	111/4
9	12	13	25/2	9	12	14	-7/2
9	12	15	-4	9	12	16	-28
9	12	17	-111	9	12	18	28
9	12	19	-4	9	13	13	127/4
9	13	14	7/2	9	13	15	4
9	13	16	4	9	13	17	-9
9	13	18	4	9	13	19	-108
9	14	14	-33/4	9	14	15	12
9	14	16	-4	9	14	17	7
9	14	18	-4	9	14	19	-4
9	15	15	-164	9	15	16	-86

a	b	c	p_{abc}	a	b	c	p_{abc}
9	15	18	344	9	16	16	-48
9	16	17	16	9	16	18	140
9	16	19	-16	9	17	17	108
9	17	18	-16	9	17	19	-16
9	18	18	-184	9	19	19	112
10	10	10	166/3	10	10	11	-24
10	10	12	-32	10	10	13	16
10	10	14	8	10	10	15	-143
10	10	16	-175	10	10	17	48
10	10	18	186	10	10	19	-16
10	11	11	-65/2	10	11	12	13
10	11	13	-58	10	11	14	-24
10	11	15	12	10	11	16	28
10	11	17	12	10	11	18	-48
10	11	19	116	10	12	12	-21/2
10	12	13	-32	10	12	14	10
10	12	15	12	10	12	16	52
10	12	17	28	10	12	18	-48
10	12	19	4	10	13	13	-107/2
10	13	14	27	10	13	15	-12
10	13	16	-28	10	13	17	28
10	13	18	8	10	13	19	116
10	14	14	-15/2	10	14	15	-12
10	14	16	-4	10	14	17	-4
10	14	18	8	10	14	19	-12
10	15	15	123	10	15	16	72
10	15	17	-16	10	15	18	-122
10	15	19	16	10	16	16	179
10	16	17	-80	10	16	18	-282
10	16	19	32	10	17	17	-48
10	17	18	48	10	17	19	-16
10	18	18	124	10	18	19	-16
10	19	19	-96	11	11	11	571/12
11	11	12	669/4	11	11	13	81/2
11	11	14	70	11	11	15	-16
11	11	16	34	11	11	17	-206
11	11	18	-6	11	11	19	-73
11	12	12	559/4	11	12	13	78
11	12	14	77	11	12	15	-50
11	12	16	-30	11	12	17	-294
11	12	18	-62	11	12	19	-44
11	13	13	-149/4	11	13	14	57/2
11	13	15	26	11	13	16	94
11	13	17	-133	11	13	18	40
11	13	19	4	11	14	14	87/4
11	14	15	-8	11	14	16	24
11	14	17	-179	11	14	18	-14
11	14	19	-66	11	15	15	12
11	15	17	52	11	15	19	-20
11	16	16	-28	11	16	17	-4
11	16	18	36	11	16	19	-144
11	17	17	151	11	17	18	100
11	17	19	92	11	18	18	-12
11	18	19	-20	11	19	19	32
12	12	12	43/4	12	12	13	55/2
12	12	14	29	12	12	15	-14
12	12	16	4	12	12	17	-132
12	12	18	-36	12	12	19	9
12	13	13	-11/4	12	13	14	43/2
12	13	16	44	12	13	17	-103
12	13	18	-46	12	13	19	2
12	14	14	-47/4	12	14	15	-2
12	14	16	-2	12	14	17	-133
12	14	18	52	12	14	19	-52
12	15	15	16	12	15	16	-8

a	b	c	p_{abc}	a	b	c	p_{abc}
12	15	17	28	12	15	18	-8
12	15	19	20	12	16	16	-36
12	16	17	-20	12	16	18	44
12	16	19	-8	12	17	17	119
12	17	18	140	12	17	19	-12
12	18	18	-8	12	18	19	20
12	19	19	12	13	13	13	7/6
13	13	14	23	13	13	15	6
13	13	16	56	13	13	17	51
13	13	18	-2	13	13	19	-15
13	14	14	33/2	13	14	15	-6
13	14	16	-30	13	14	17	-32
13	14	18	2	13	14	19	-28
13	15	15	4	13	15	16	16
13	15	17	-20	13	15	18	-16
13	15	19	-12	13	16	16	-4
13	16	17	-12	13	16	18	-12
13	16	19	-168	13	17	17	137
13	17	18	4	13	17	19	-76
13	18	18	12	13	18	19	4
13	19	19	16	14	14	14	-52/3
14	14	16	6	14	14	17	9
14	14	19	-1	14	15	15	8
14	15	16	8	14	15	17	4
14	15	18	-24	14	15	19	12
14	16	16	-12	14	16	17	4
14	16	18	-4	14	16	19	16
14	17	17	137	14	17	18	-36
14	17	19	92	14	18	18	16
14	18	19	-4	14	19	19	28
15	15	15	-68/3	15	15	16	4
15	15	17	-32	15	15	19	-8
15	16	16	4	15	16	18	-8
15	16	19	-16	15	17	18	32
15	18	19	16	16	16	16	-116/3
16	16	17	32	16	16	18	88
16	17	17	48	16	17	18	-32
16	18	18	-88	16	18	19	16
16	19	19	136	17	17	17	64/3
17	17	18	-136	17	17	19	-32
17	19	19	16	18	18	18	64/3
18	18	19	-8	19	19	19	-8/3

Data completeness. The three tables contain 1,677 + 24 + 1,414 coefficient entries. No coefficient is defined implicitly by a computer search. Their exact identities can be verified from these tables, the 24 specified algebraic points, and the formulas in the body of the paper.